

Druggability Predictions of Ribonucleic Acid Crystal Structures

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Do we really need new drug targets?

Antibiotic crisis

New antiviral drugs

Cancer and rare diseases

Large unexplored territory

Human genome: 30.000 genes

FDA-approved drugs scope: 667 targets

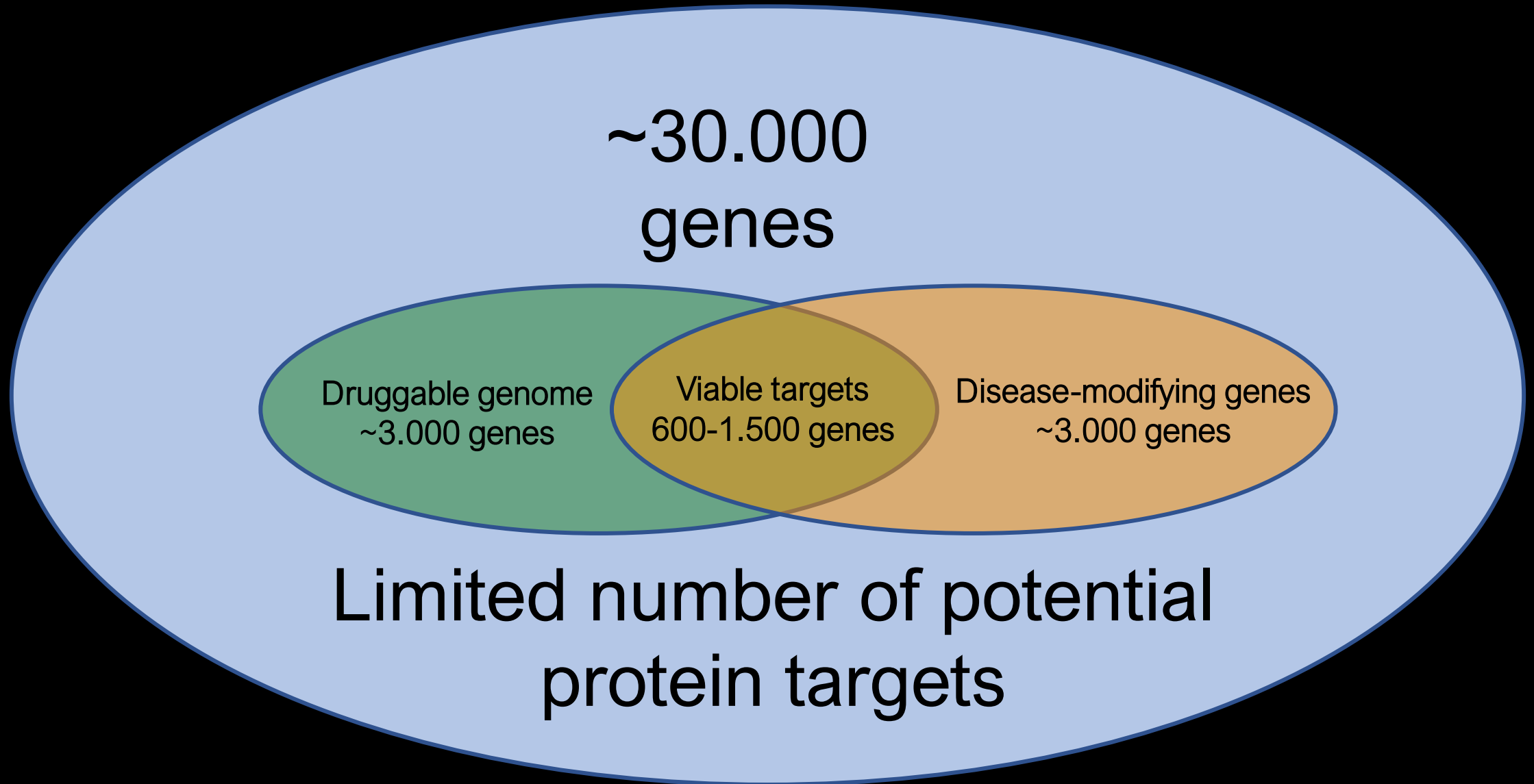
Lushniak, B. D. *Public Health Rep* **2014**, 129, 314-6.

Aslam, *et al. Infect Drug Resist* **2018**, 11, 1645-1658.

Asahchop, E. L.; *et al. Antimicrob Agents Ch* **2012**, 56, 5000-5008.

Santos, R.; Ursu, O. *et al. Nat Rev Drug Discov* **2017**, 16, 19-34.

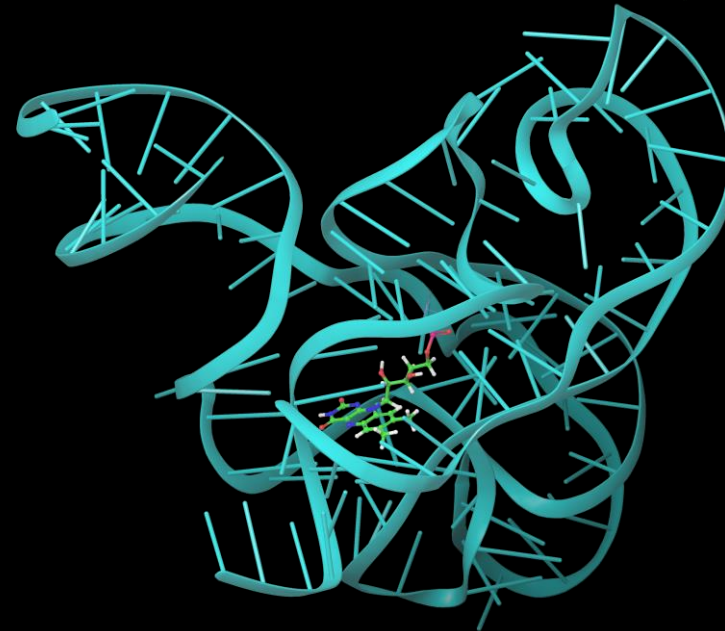
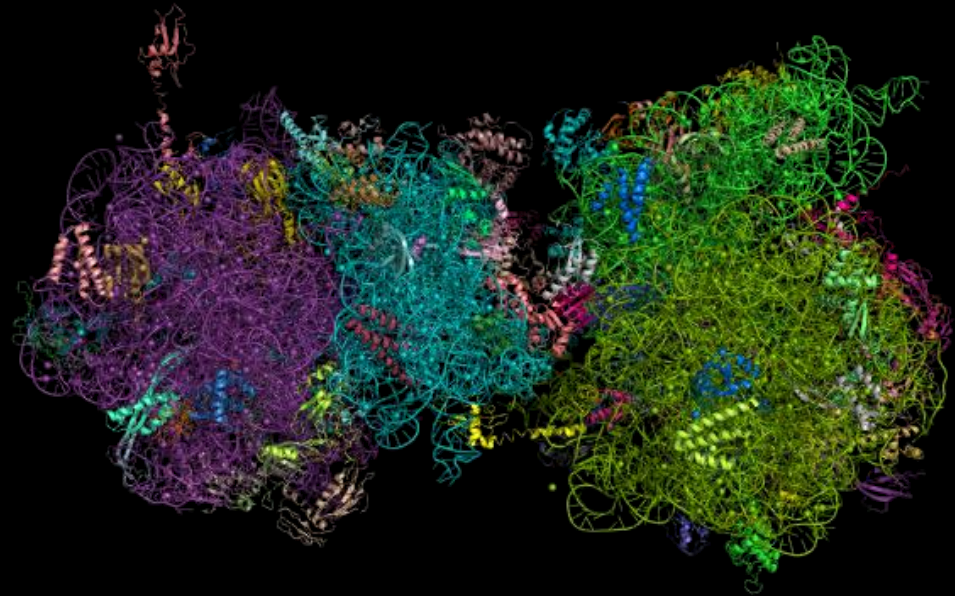
The human genome:



Which drug targets exist outside of the protein domain?

RNA as a drug target

- Folded RNA can create binding pockets
- Small molecules
- Examples:
Ribosomes
Riboswitches



FMN riboswitch (2yie)

Hermann, T. *Curr Opin Struc Biol* **2005**, 15, 355-366.

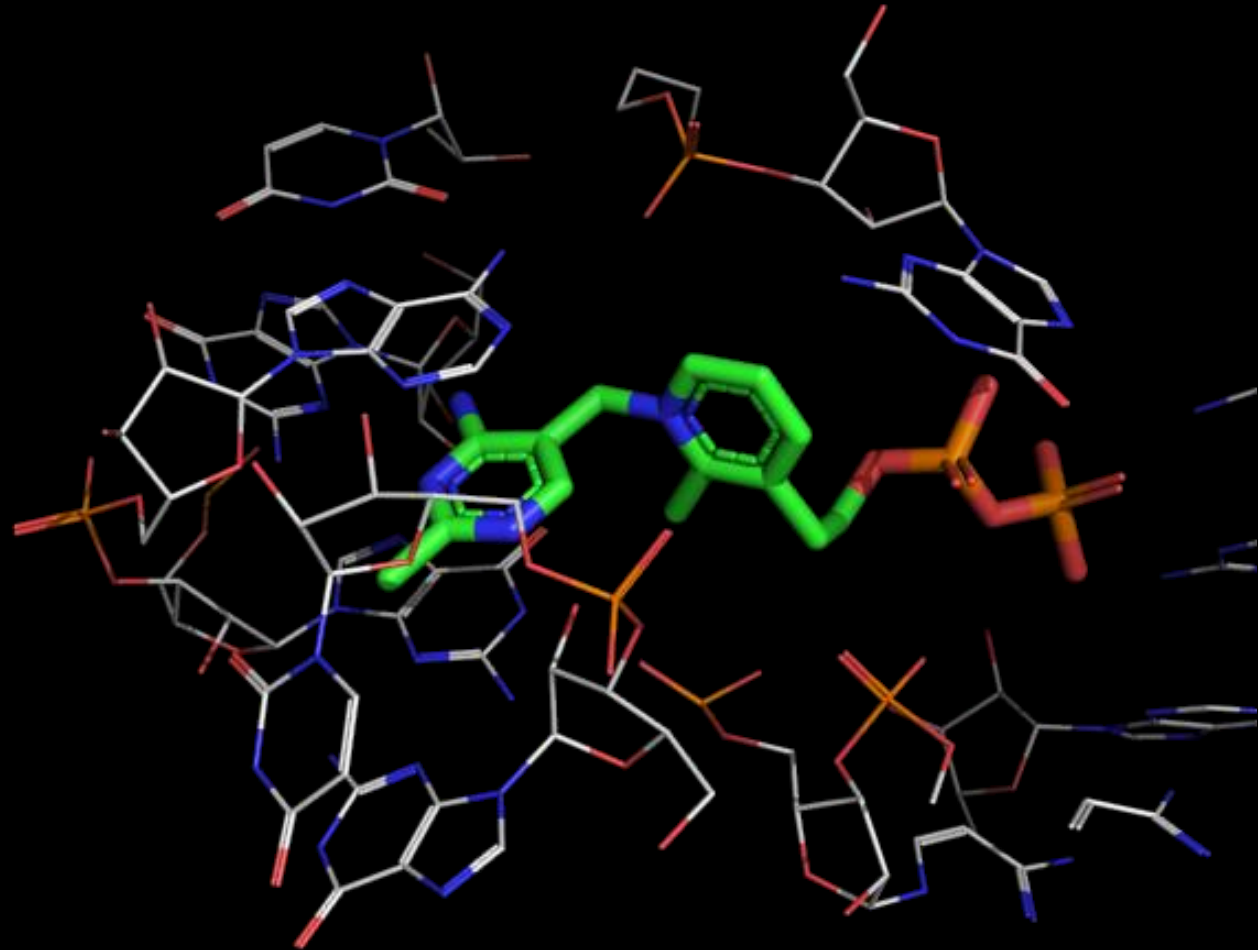
Pearson, N. D.; Prescott, C. D. *Chem Biol* **1997**, 4, 409-414.

Rekand, I. H.; Brenk, R.. *Future Medicinal Chemistry* **2017**, 9, 1649-1662.

Serganov, A.; Nudler, E. *Cell* **2013**, 152, 17-24.

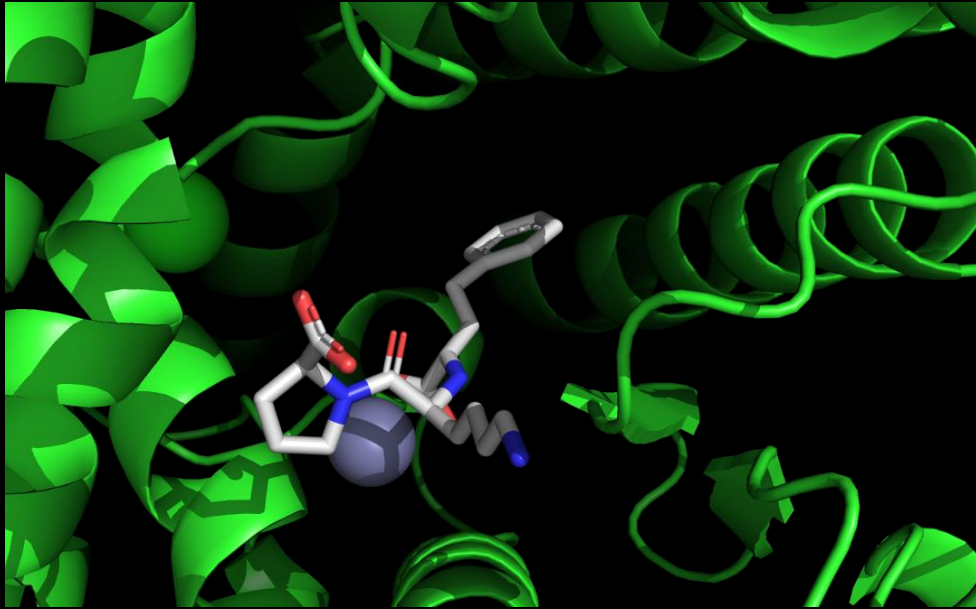
But isn't RNA...

- “Too unstable?”
Protein database (PDB):
Containing RNA: 4700
Only RNA: 1400
- “Too polar?”
Nucleic bases provide
hydrophobic surface

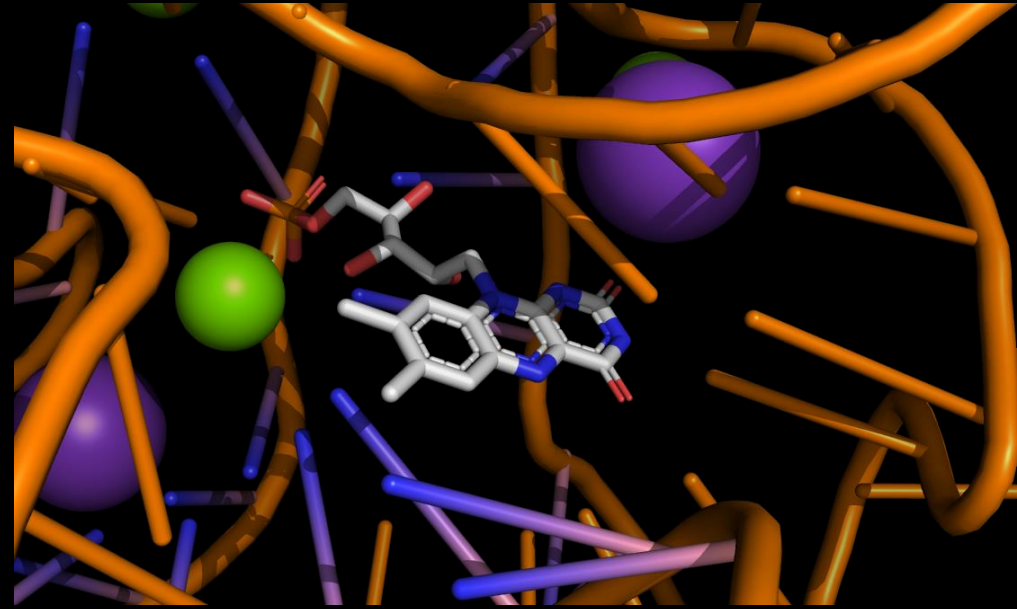


TPP riboswitch (PDB code: 3d2v)

Protein vs. RNA – what to consider?



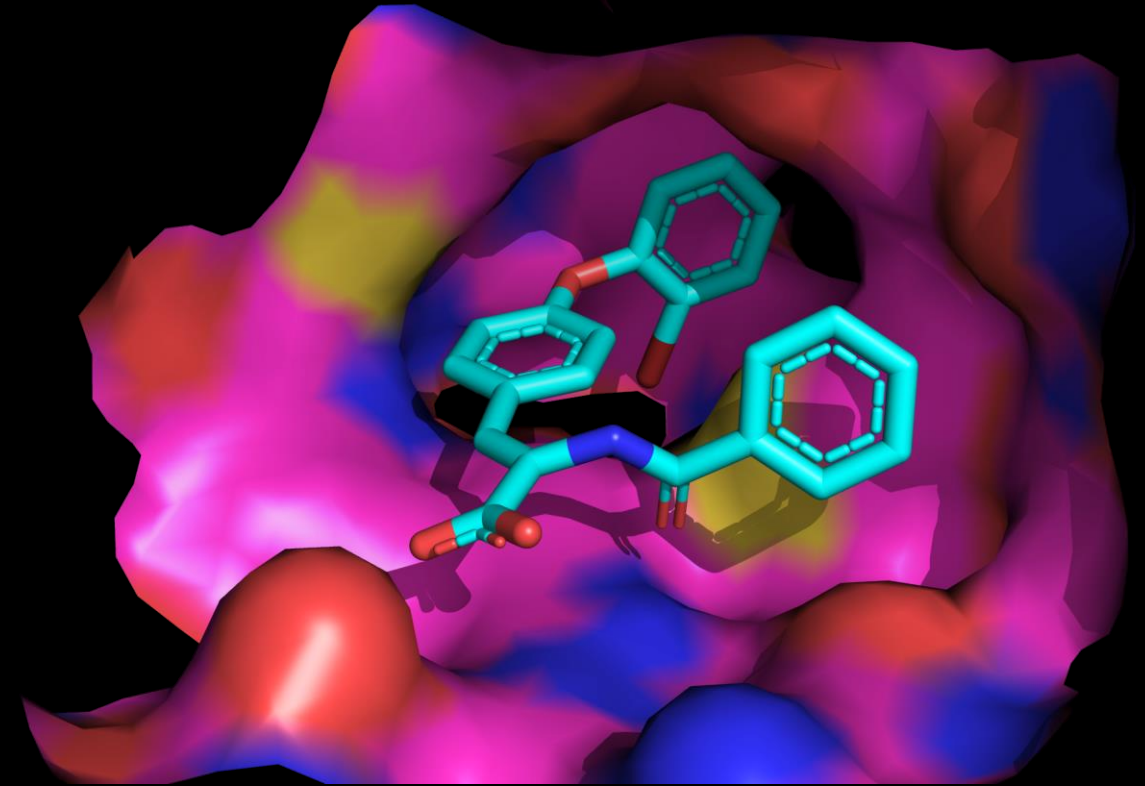
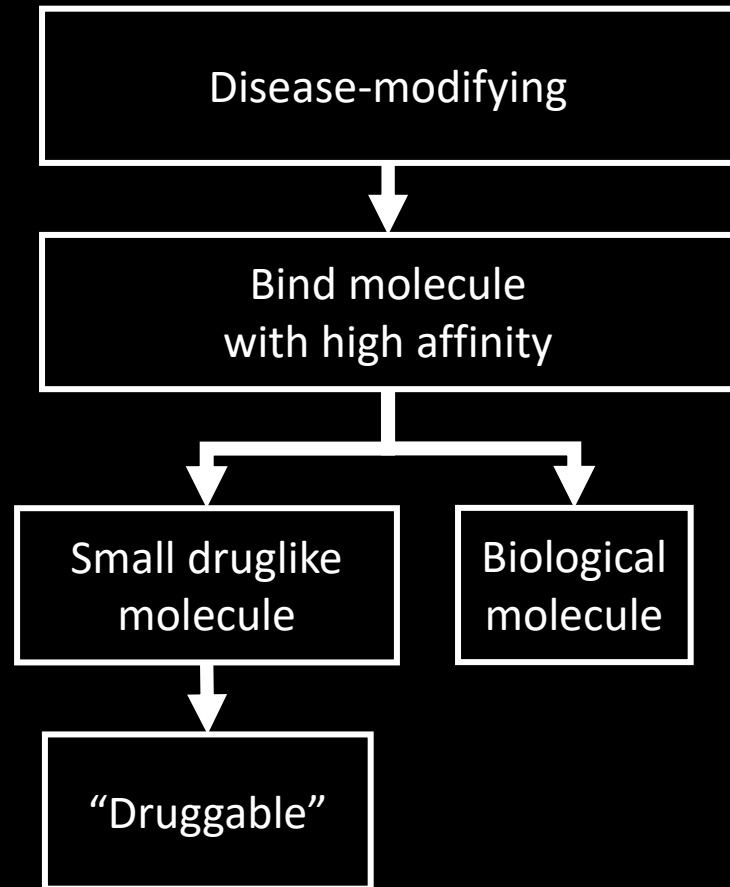
Human Angiotensin Converting Enzyme in complex with lisinopril. (1o86)



FMN riboswitch in complex with FMN (2yie)

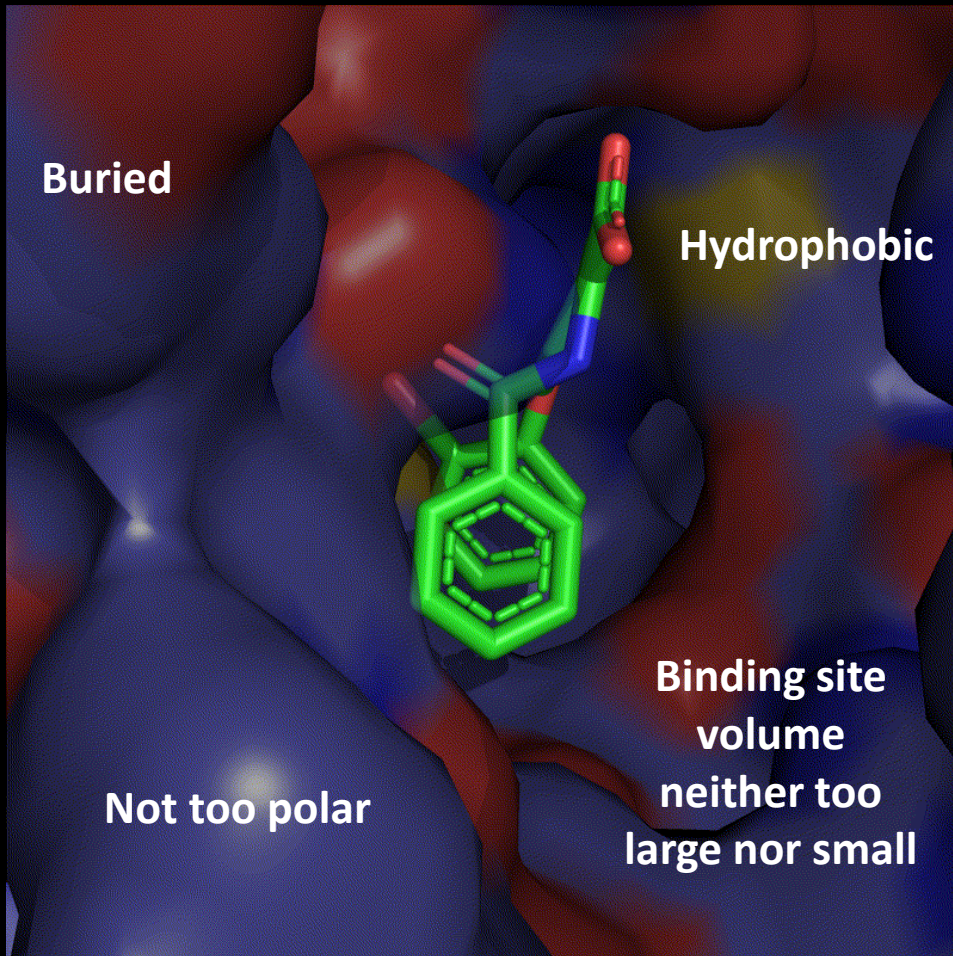
- 20 amino acids vs. 4 residues
- Basic interactions principles apply

What criteria should we ask from a drug target?



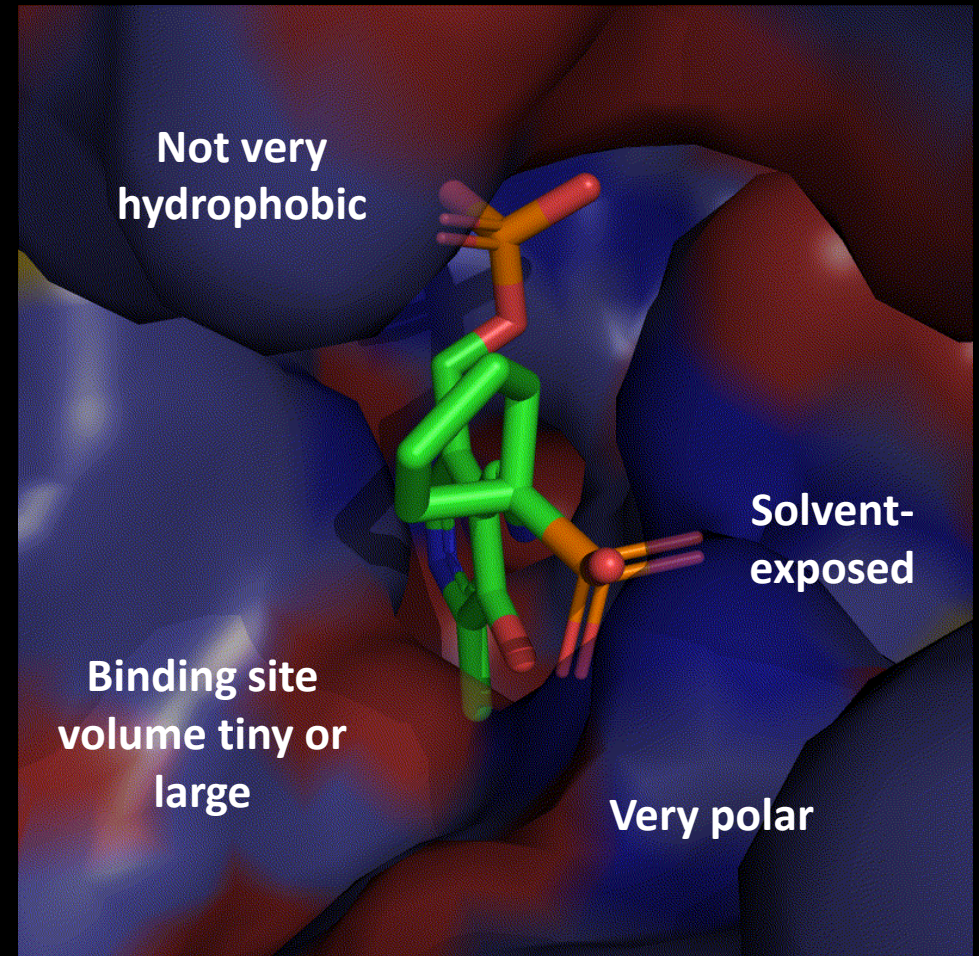
Hepatitis C virus RNA polymerase complex with inhibitor (PDB: 1yvf)

Druggable



Hepatitis C virus RNA polymerase complex with inhibitor (PDB: 1yvf)

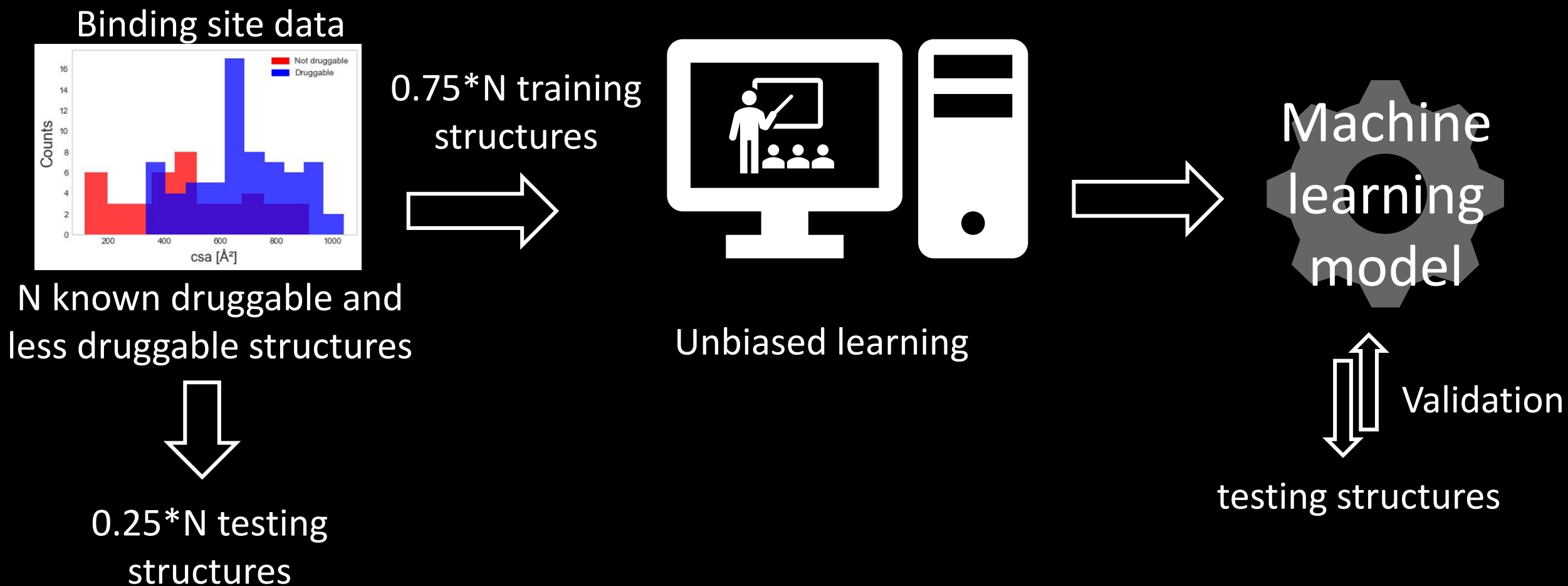
Less druggable



Dialkylglycine decarboxylase complex with inhibitor (PDB: 1m0n)

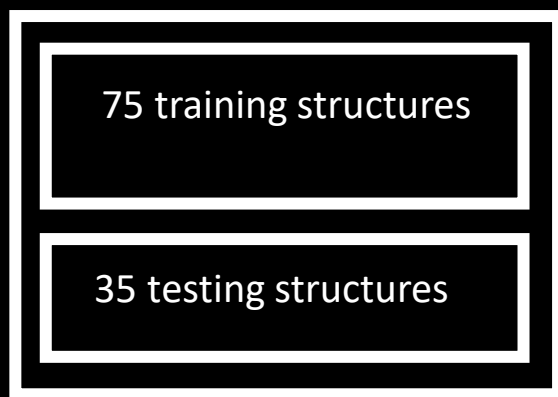
How can we predict if a binding site is druggable?

A word on machine learning

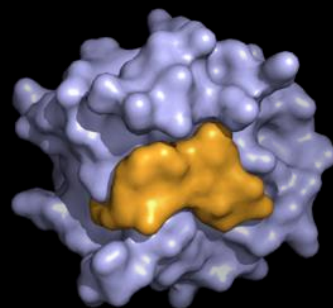
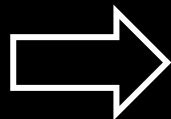


DrugPred: a druggability prediction software

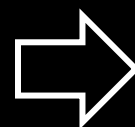
110 non-redundant known druggable or less druggable protein structures



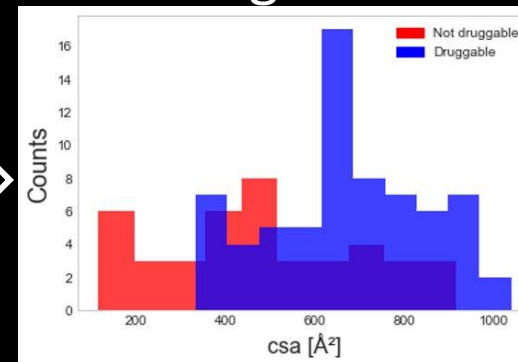
Data preparation



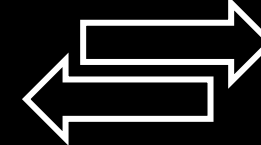
Calculate binding site properties



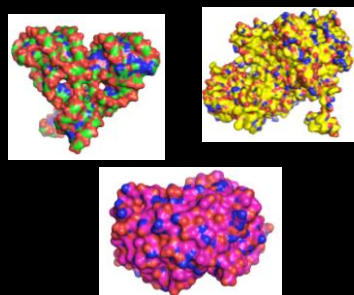
Binding site data



Evaluate and improve

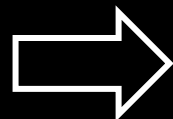


Machine learning model



Retrieve crystal structures with ligands

Data preparation



Calculate binding site properties

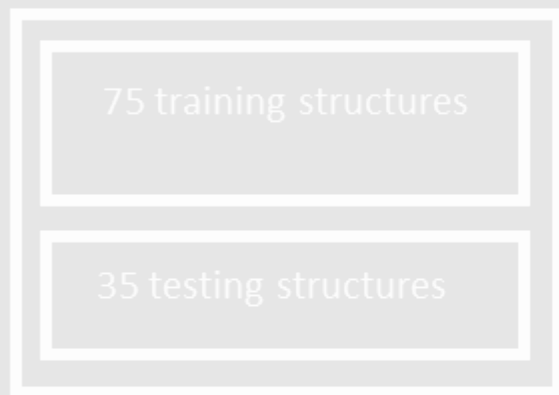


Machine learning model



DrugPred: a druggability prediction software

110 non-redundant known druggable or less druggable protein structures

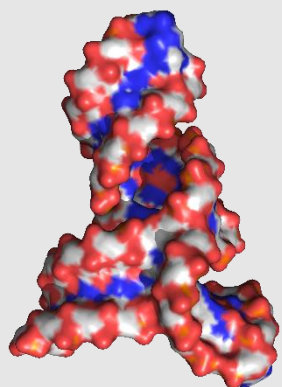


Binding site data

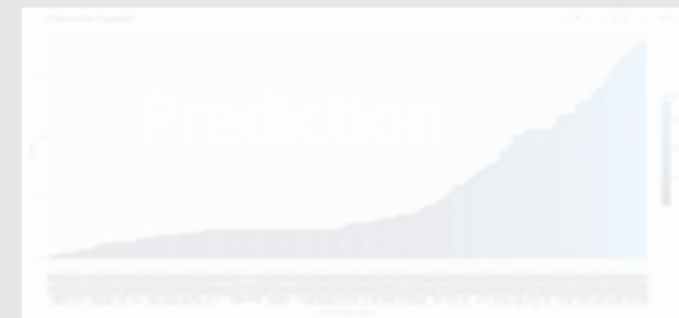
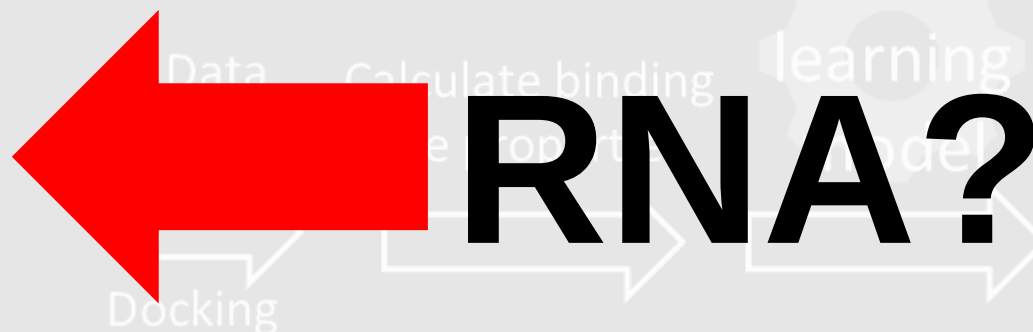


Evaluate and improve

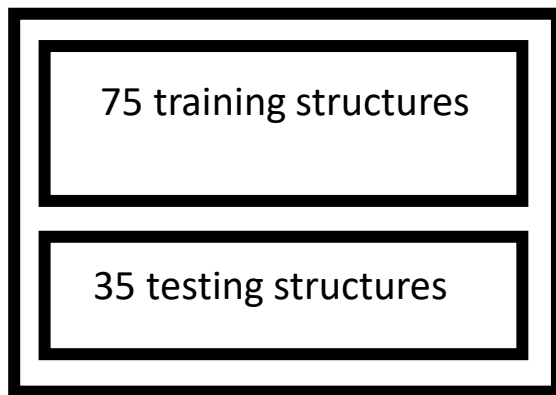
Machine learning model



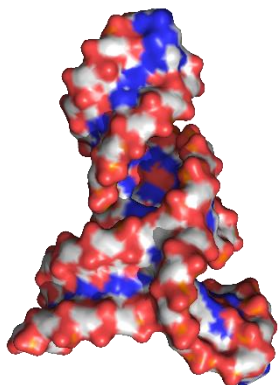
Retrieve crystal structures with ligands



110 non-redundant known druggable or
less druggable protein structures



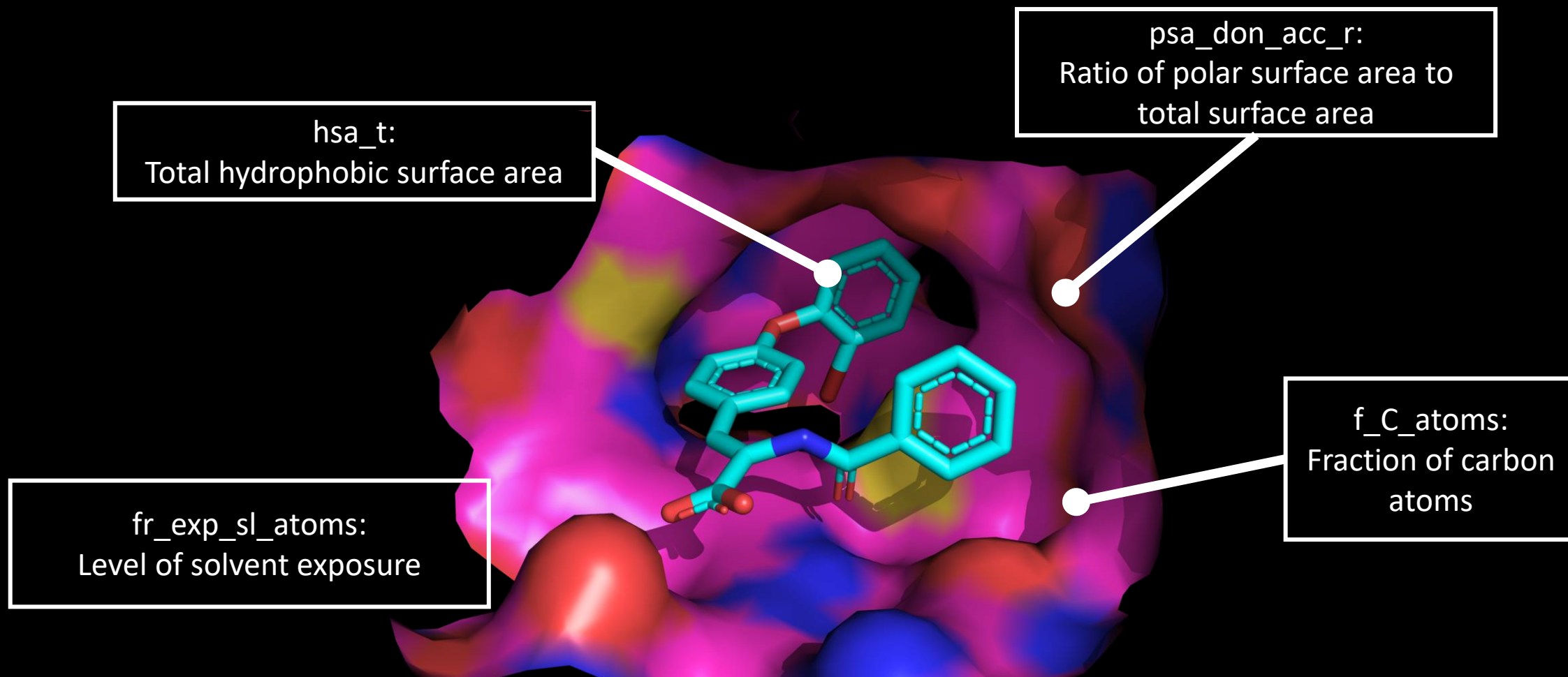
Protein set



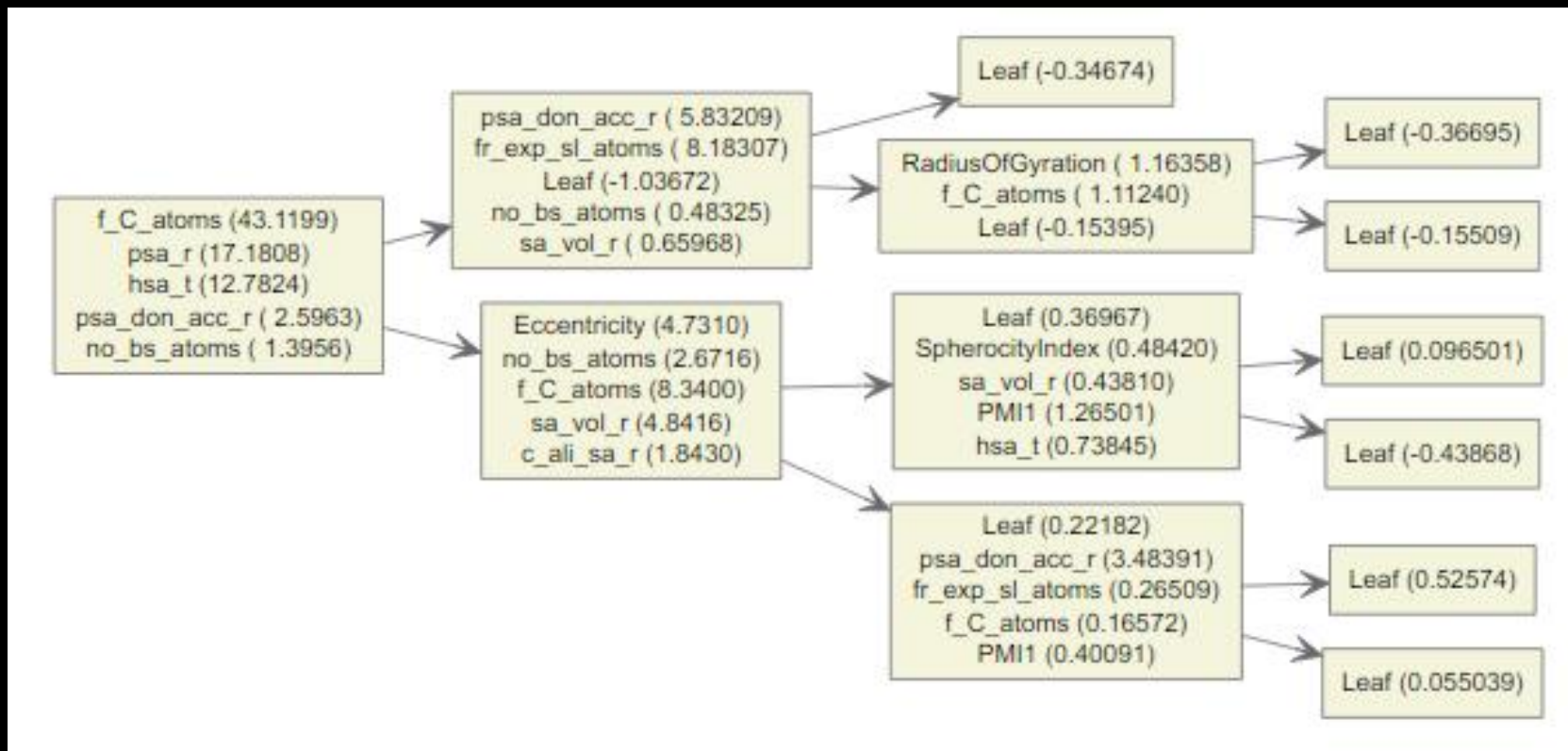
The challenge:

- **Find common descriptors**
 - **High performance**

Some binding site descriptor examples



Decision tree-based machine learning



So we gave it a shot

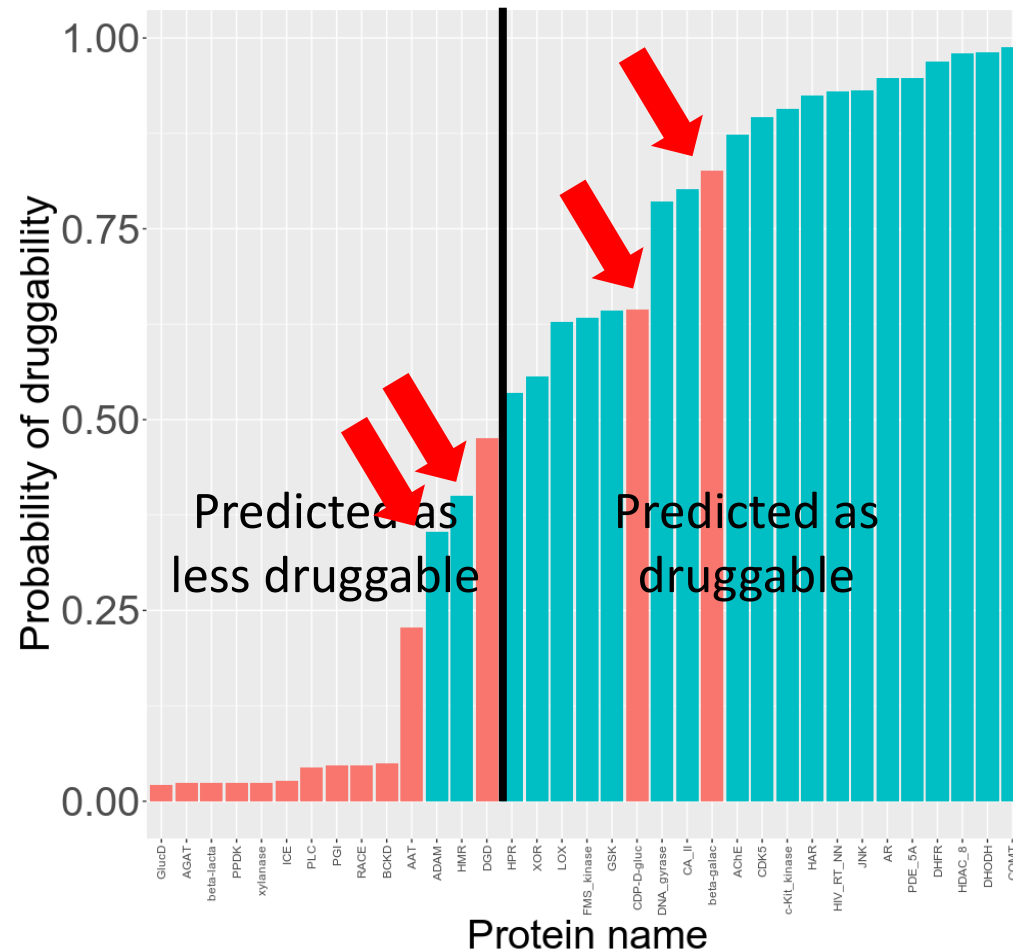
New model

99% callback accuracy

New model

89% callback accuracy

Class ■ Less Druggable ■ Druggable

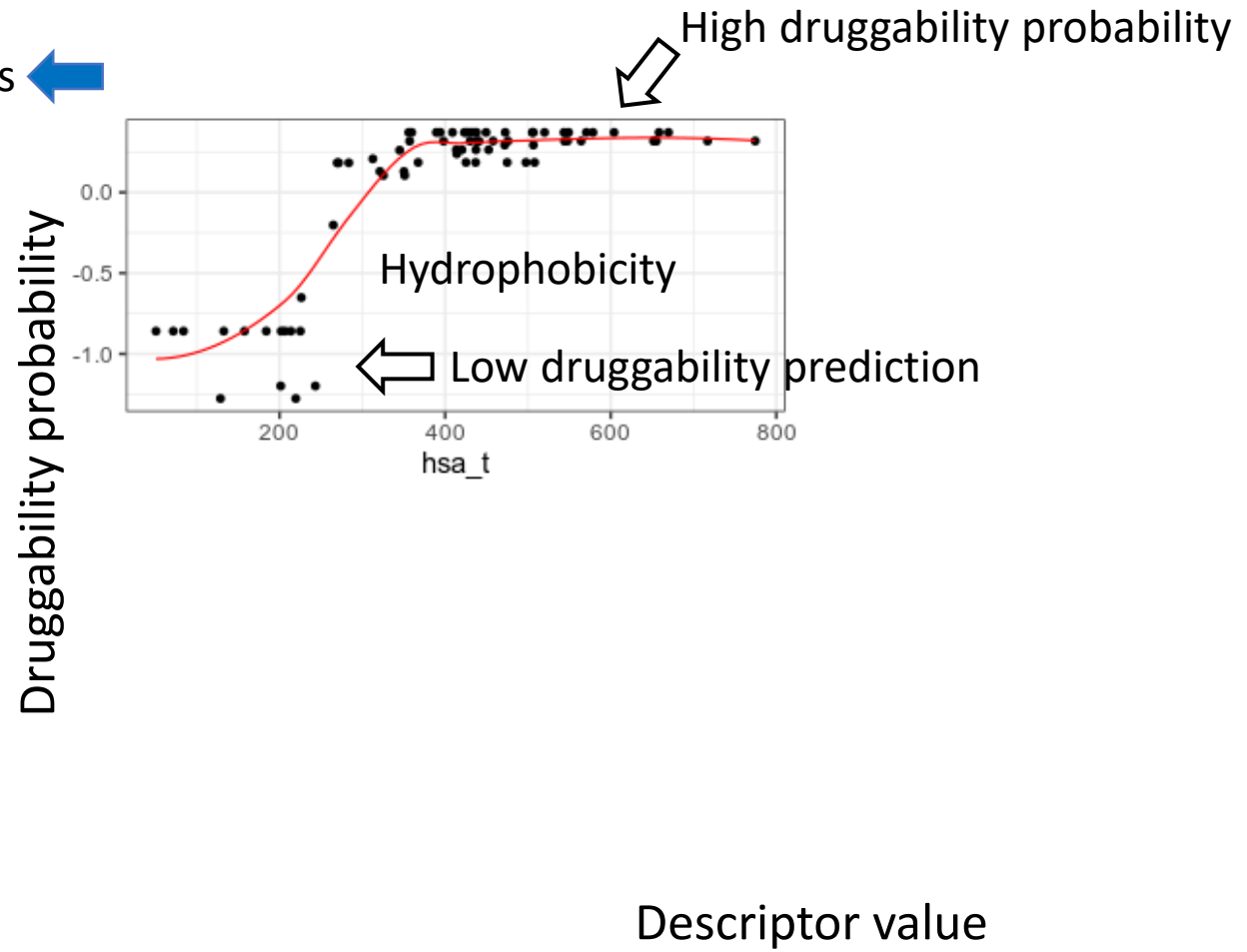
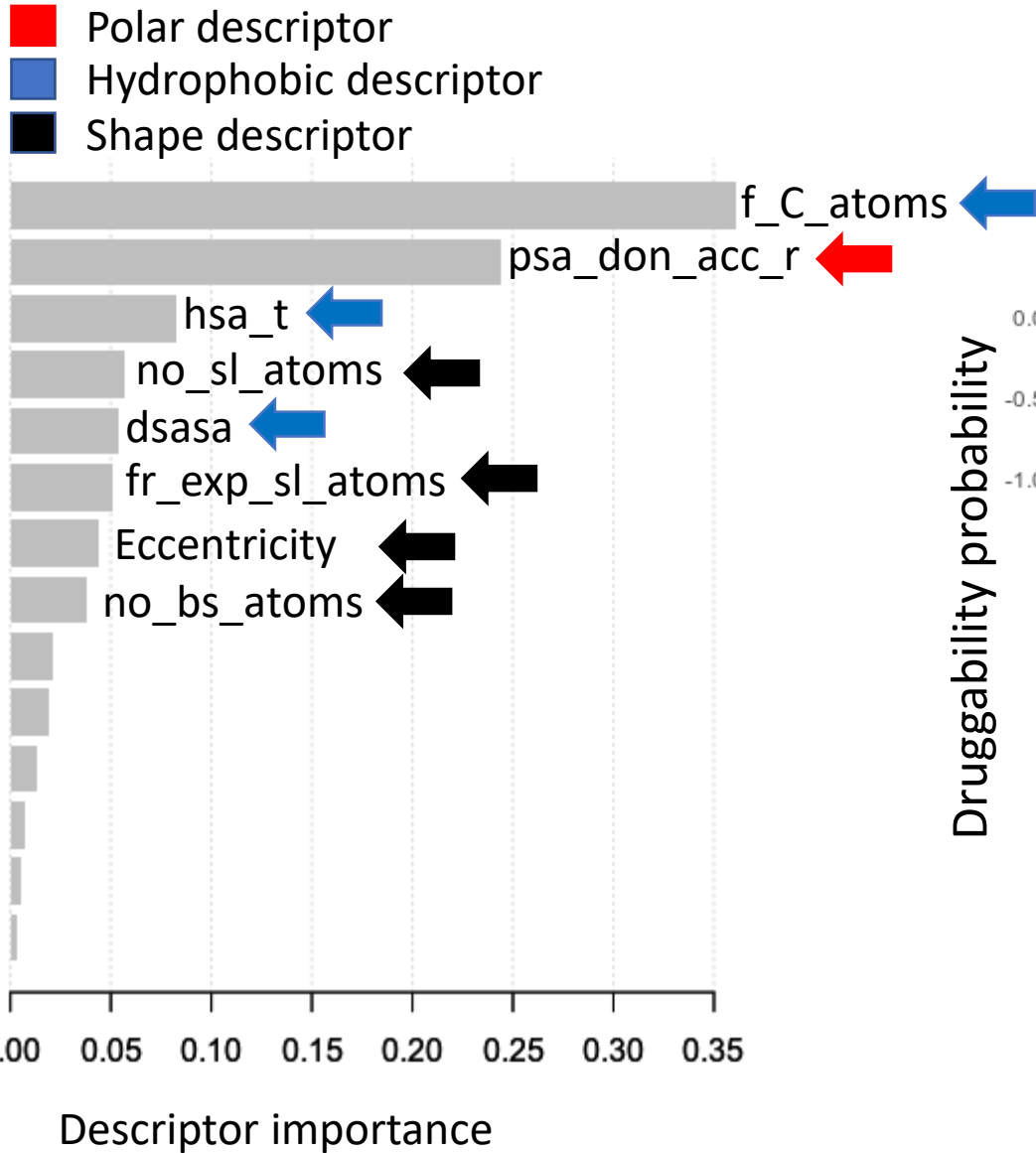


Performance new vs. old model

	Training [druggable / less druggable]		Testing [druggable / less druggable]	
	Drugpred_RNA	DrugPred 2.0	Drugpred_RNA	DrugPred 2.0
Accuracy	0.99	0.91	0.89	0.94
Precision	1.00 / 0.97	0.92 / 0.89	0.90 / 0.85	0.95 / 0.93
Recall	0.98 / 1.00	0.94 / 0.86	0.90 / 0.85	0.95 / 0.93

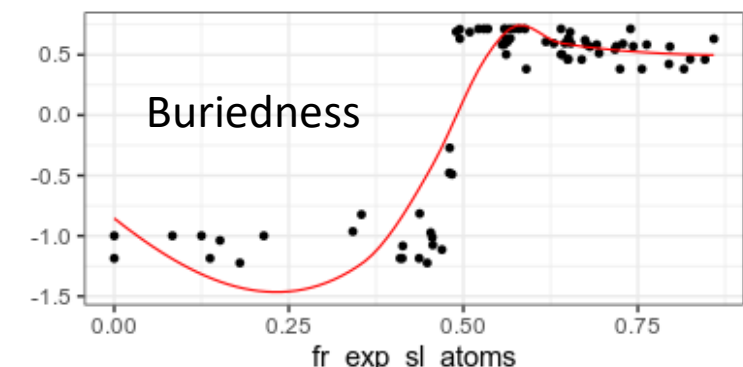
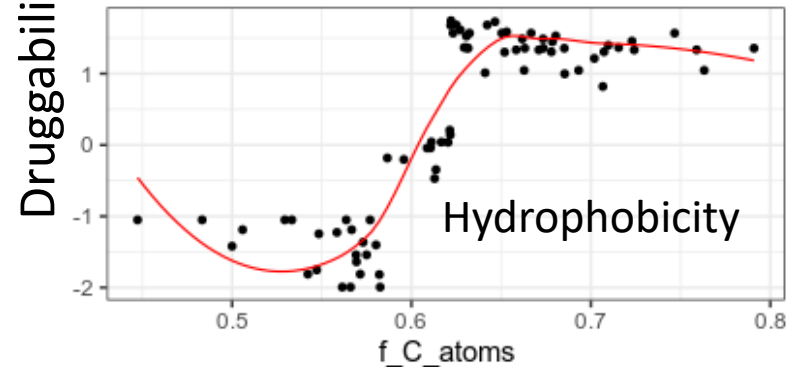
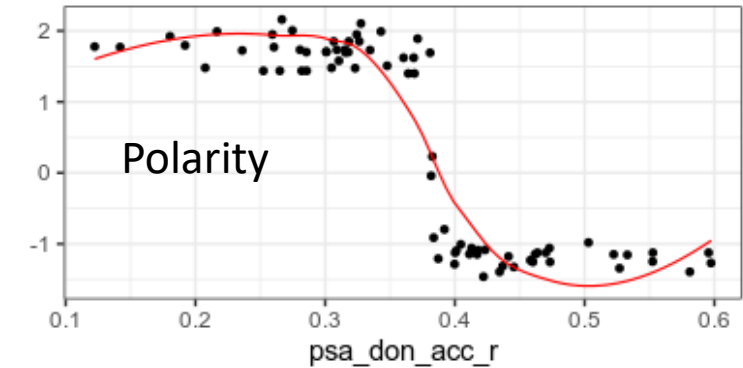
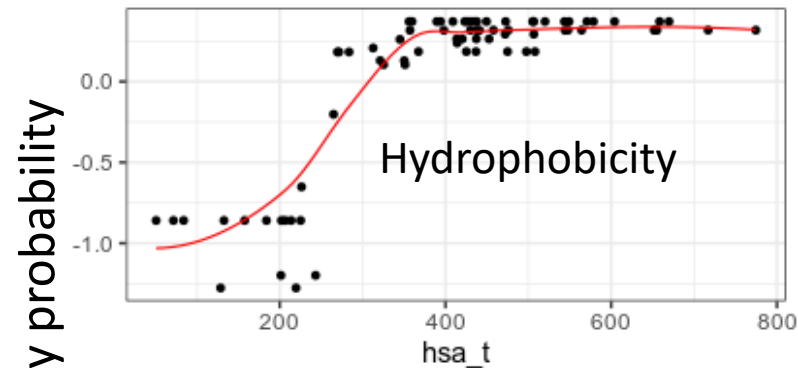
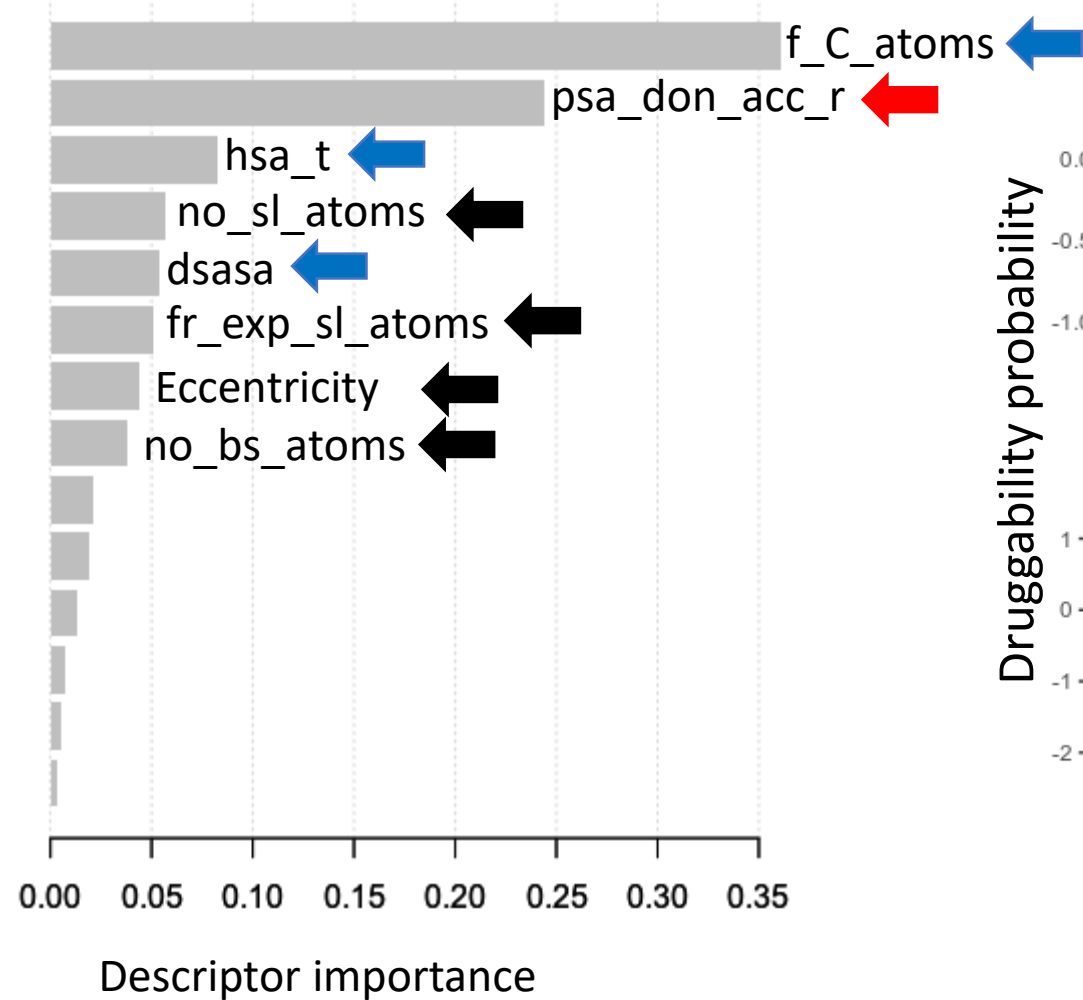
Similar performance

Which descriptors matter?



Which descriptors matter?

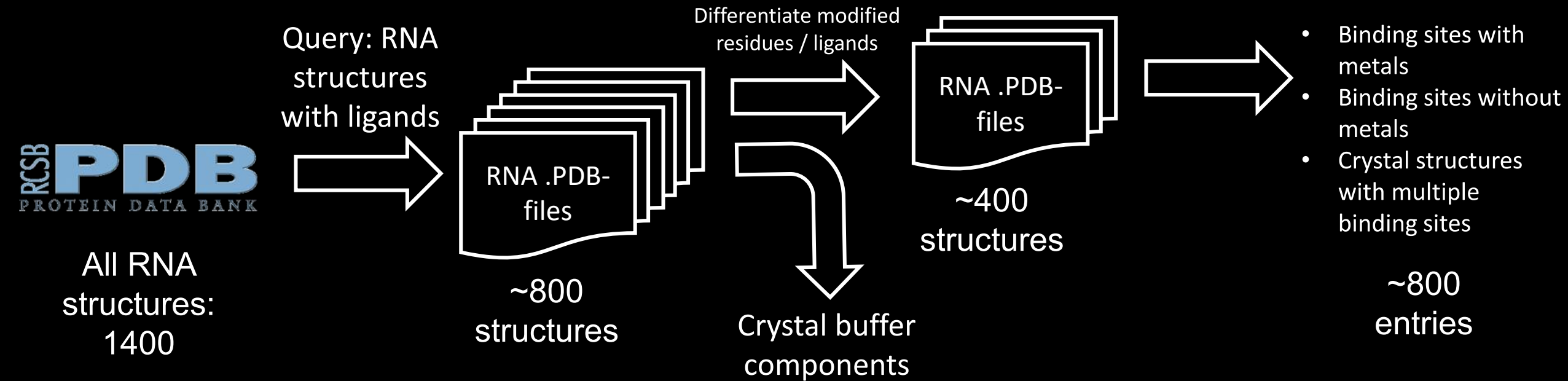
- Polar descriptor
- Hydrophobic descriptor
- Shape descriptor



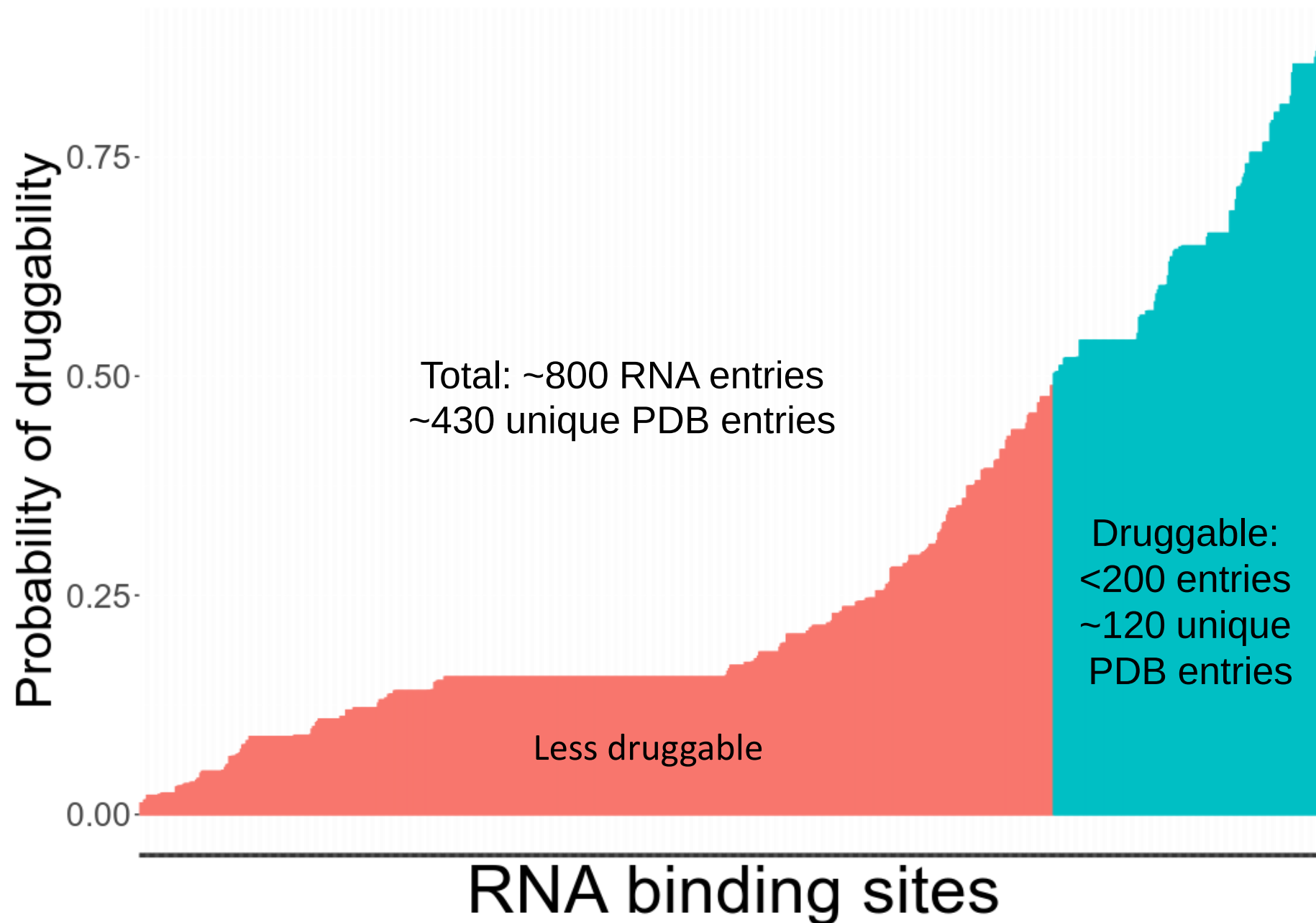
Descriptor value

Then we moved on to RNA

Creating our RNA data set

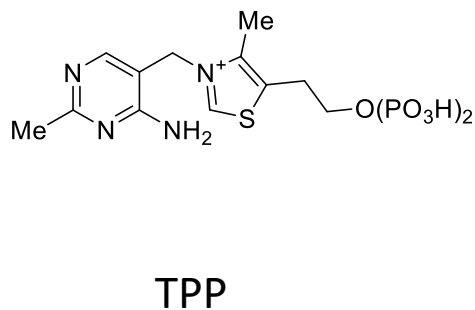
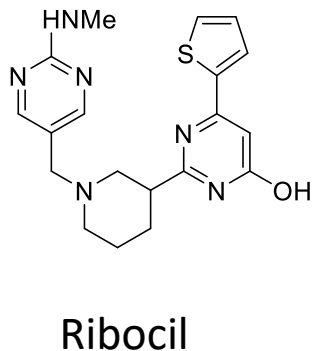
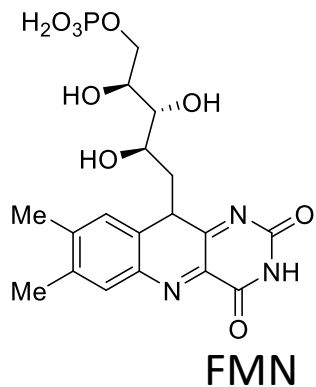
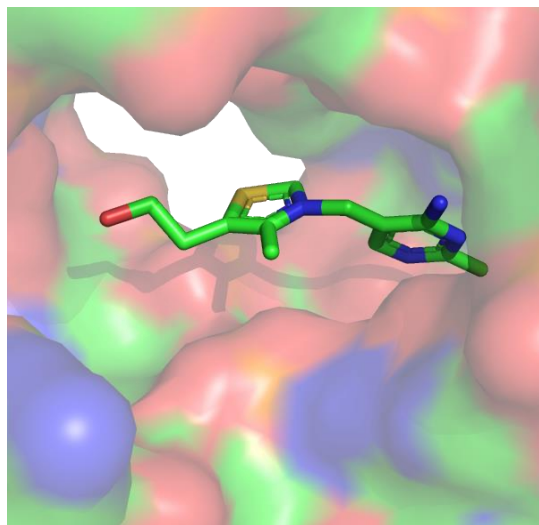
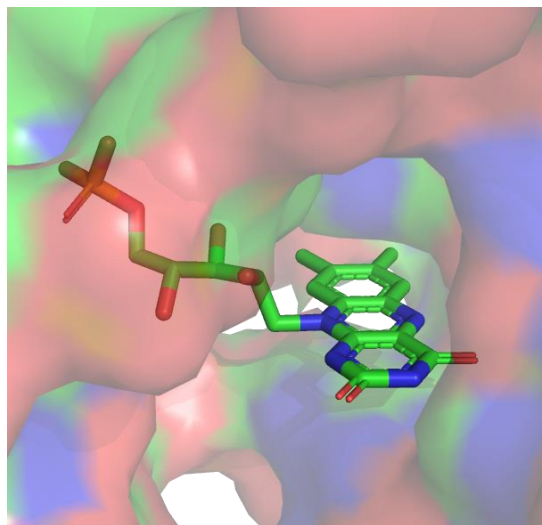


Predicted RNA druggability

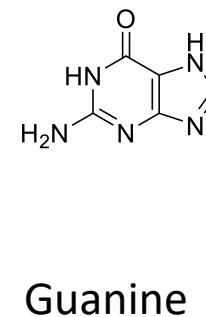
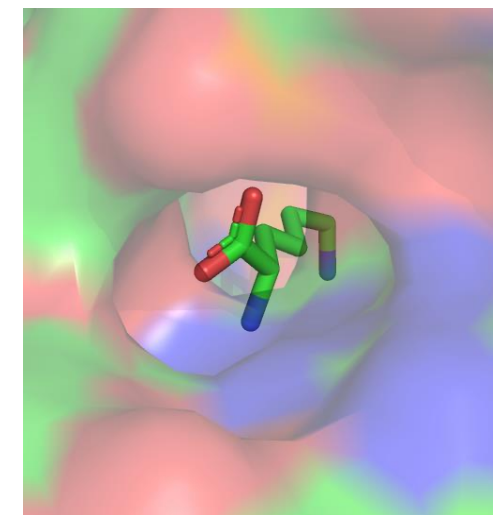
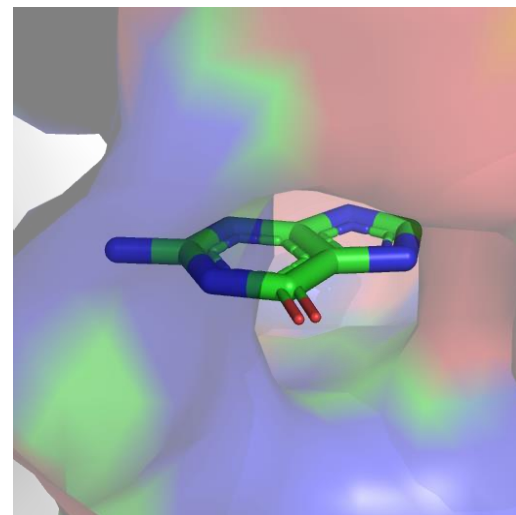


How do we evaluate our RNA predictions?

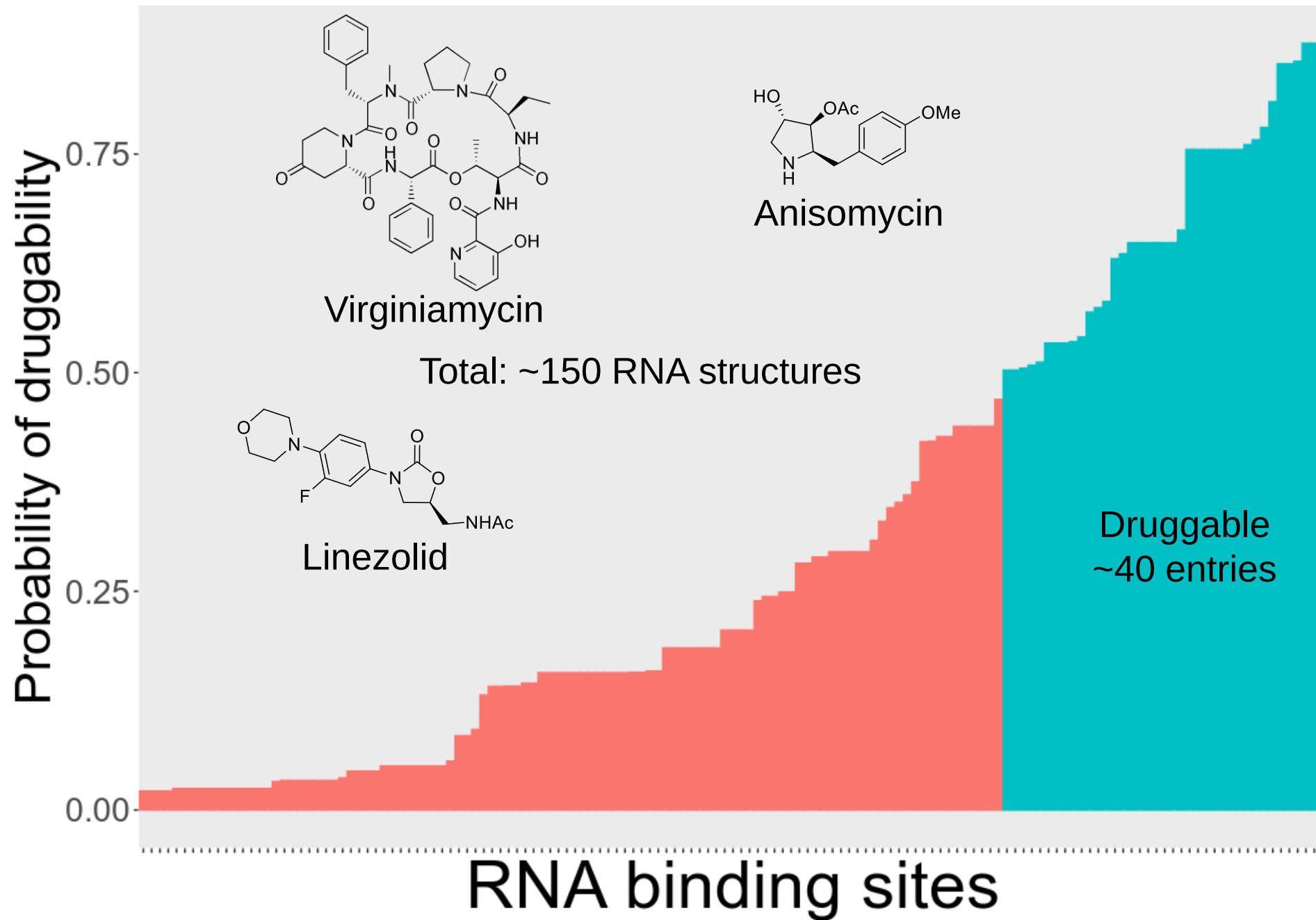
Predicted as
druggable



Predicted as
less druggable



Predicted ribosome druggability

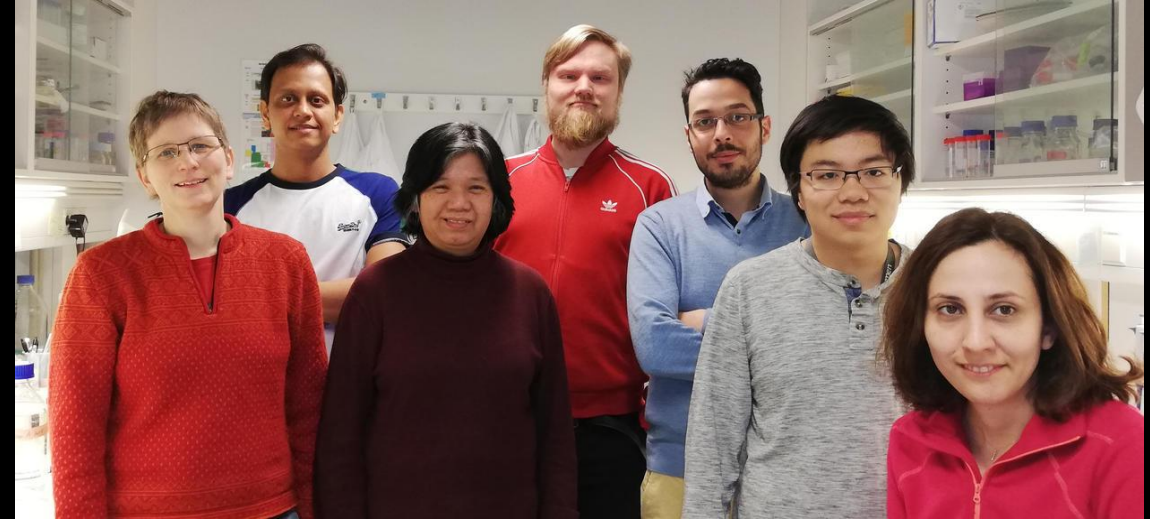


Conclusion

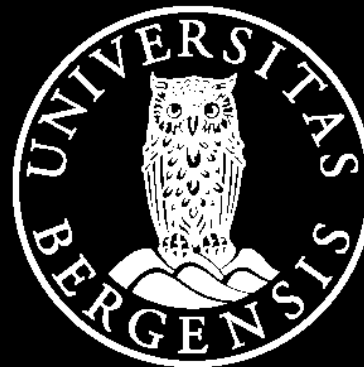
- Successfully created a druggability prediction method
- Compliant with RNA
- High accuracy
- Model behaviour on both protein testing set and RNA sample set as expected

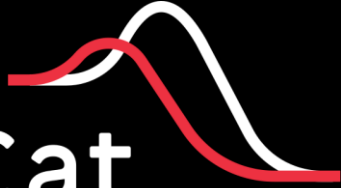
Thanks for your attention!

- Ruth Brenk and other members of my group
- Sturla Magnus Grøndal
- Helpful and ready minds of Biomedicine
- Sigma2



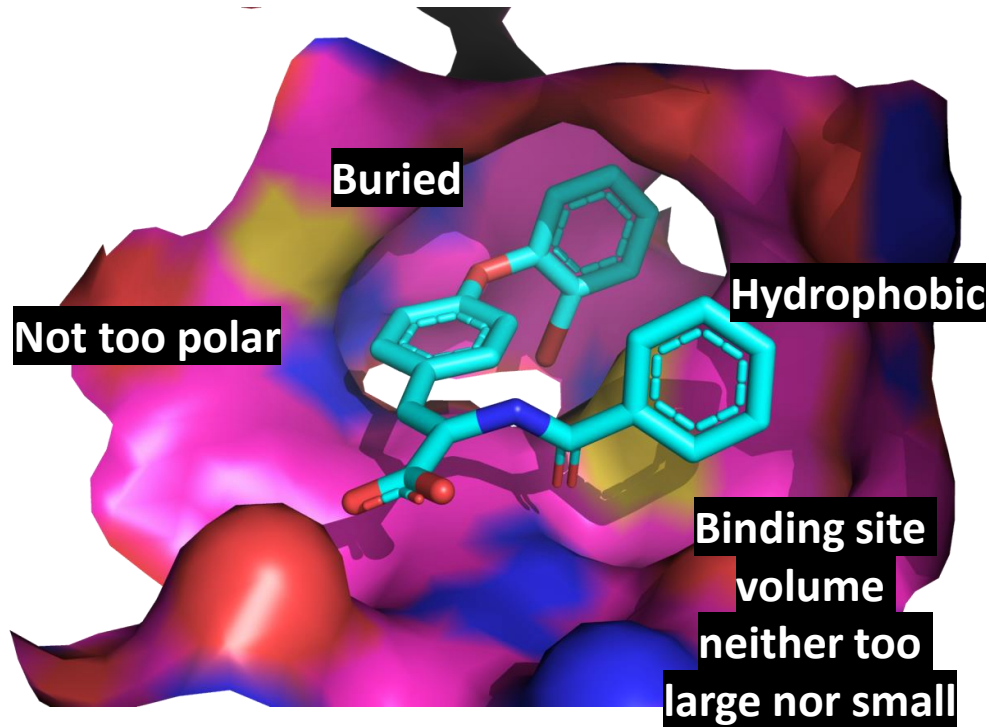
UNINETT
sigma2



BioCat 
Norwegian Graduate School
in Biocatalysis

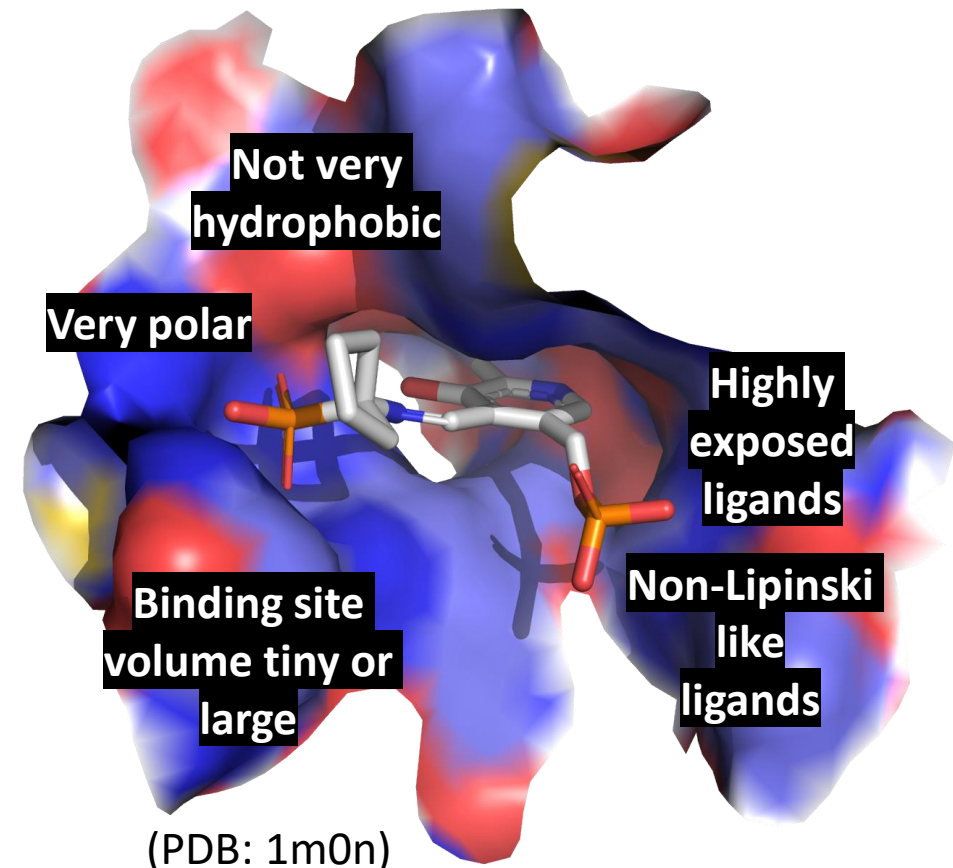
What if we generalize binding site properties?

Druggable

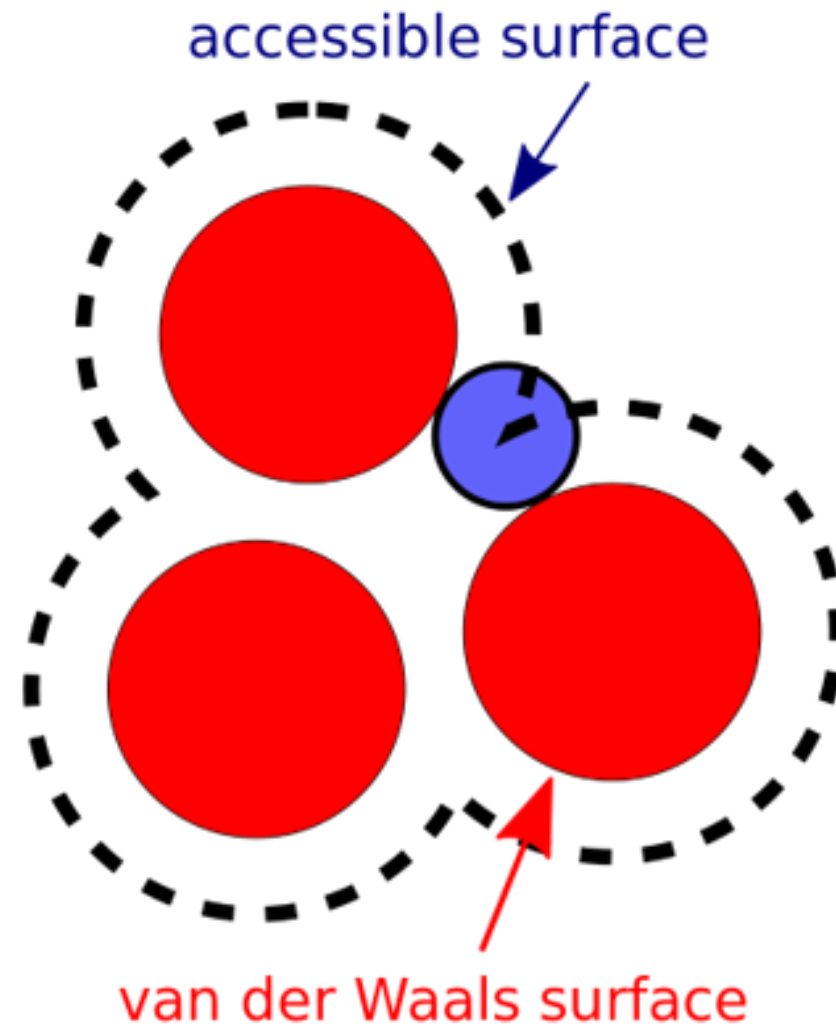


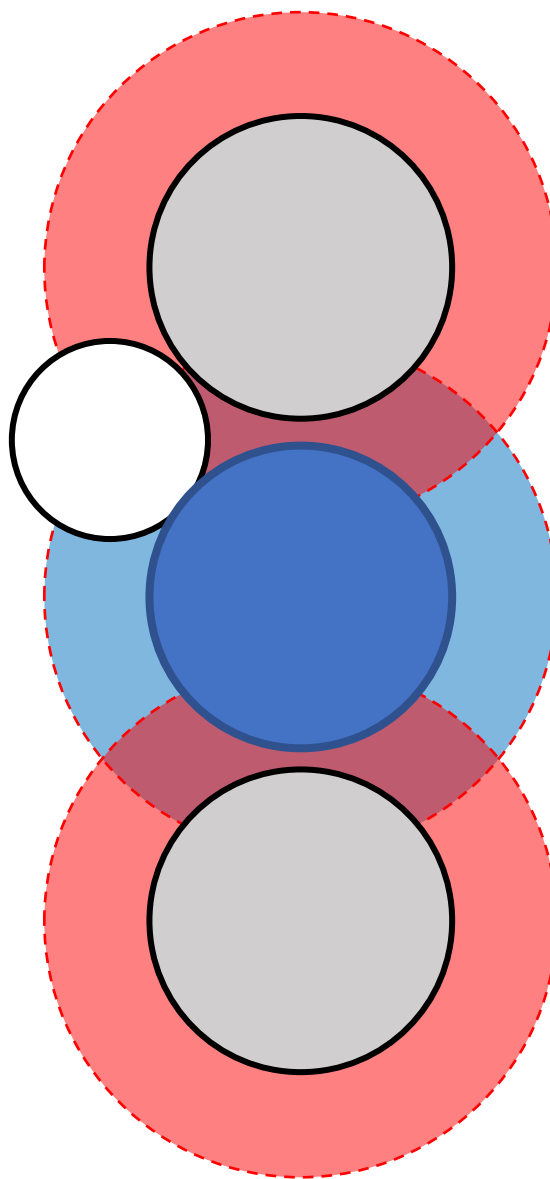
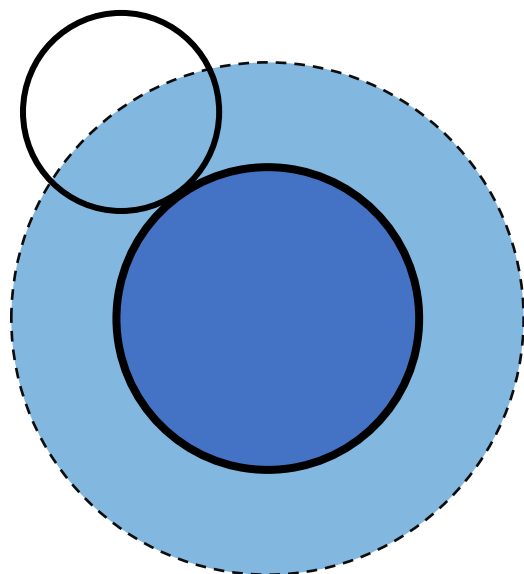
(PDB: 1yvf)

Less druggable



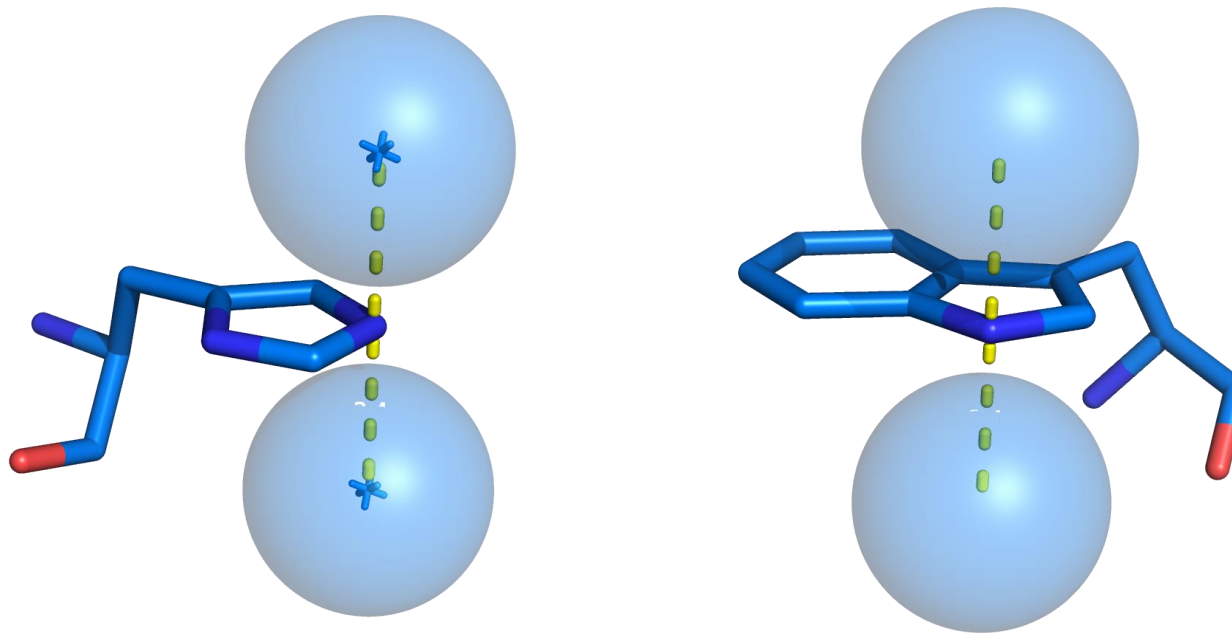
(PDB: 1m0n)





Why do we need to block aromatic N-atoms?

- Lone pair electrons
- Deny surface area (SA) for polar contribution
- + Hydrophobic SA
- - Polar SA
- Add "dummy" carbon atoms
- Dummy atoms SA is not included



Blocked aromatic *N*-atoms

Method

- Does an atom need to be blocked?
- Find unit vector (u)
 $AB \times AC = u$
- Scale (us), and adjust the unit vector to desired dummy atom coordinates

(D_{xyz}):

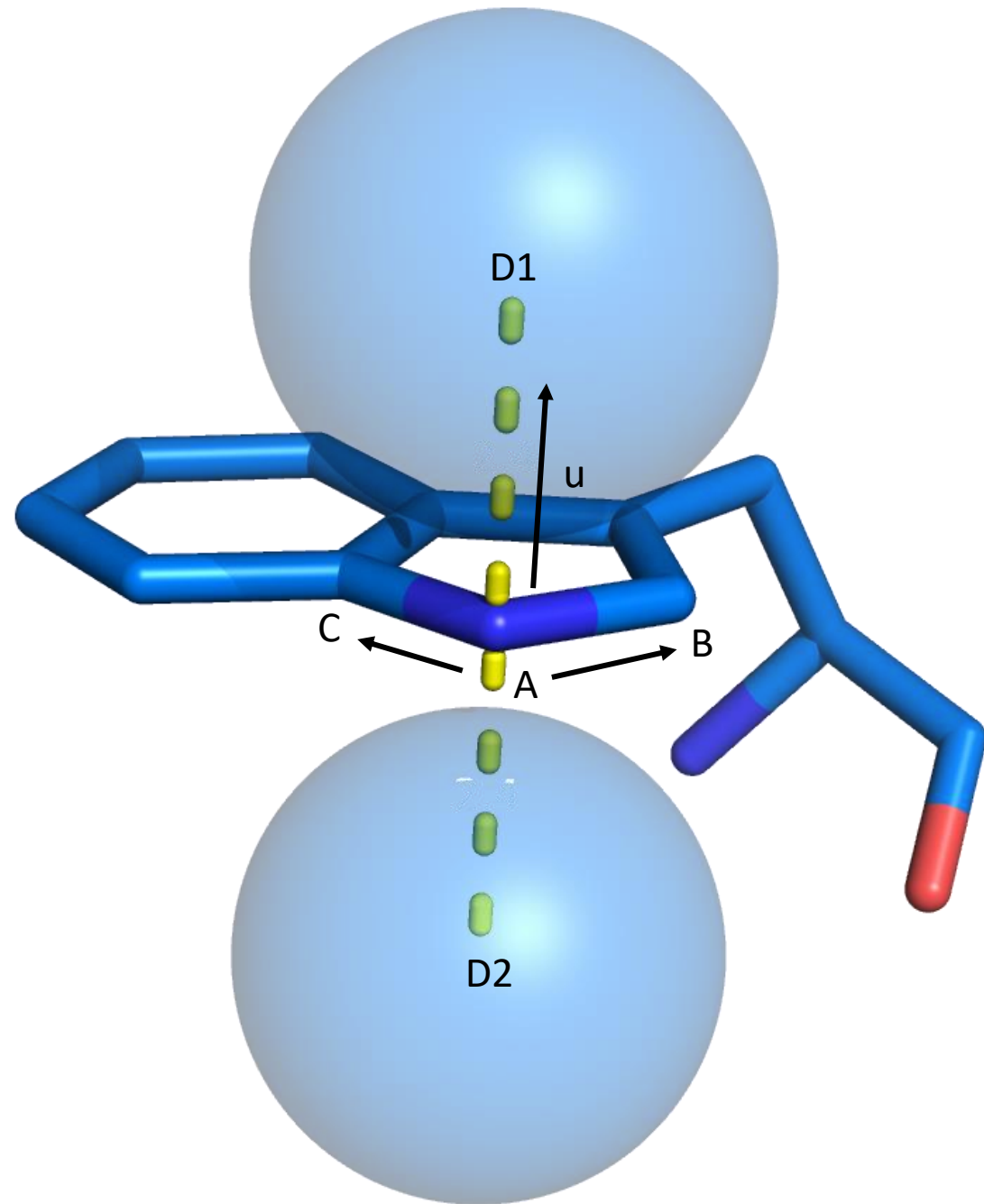
$$D1_{xyz} = us_{xyz} - N_{xyz}$$

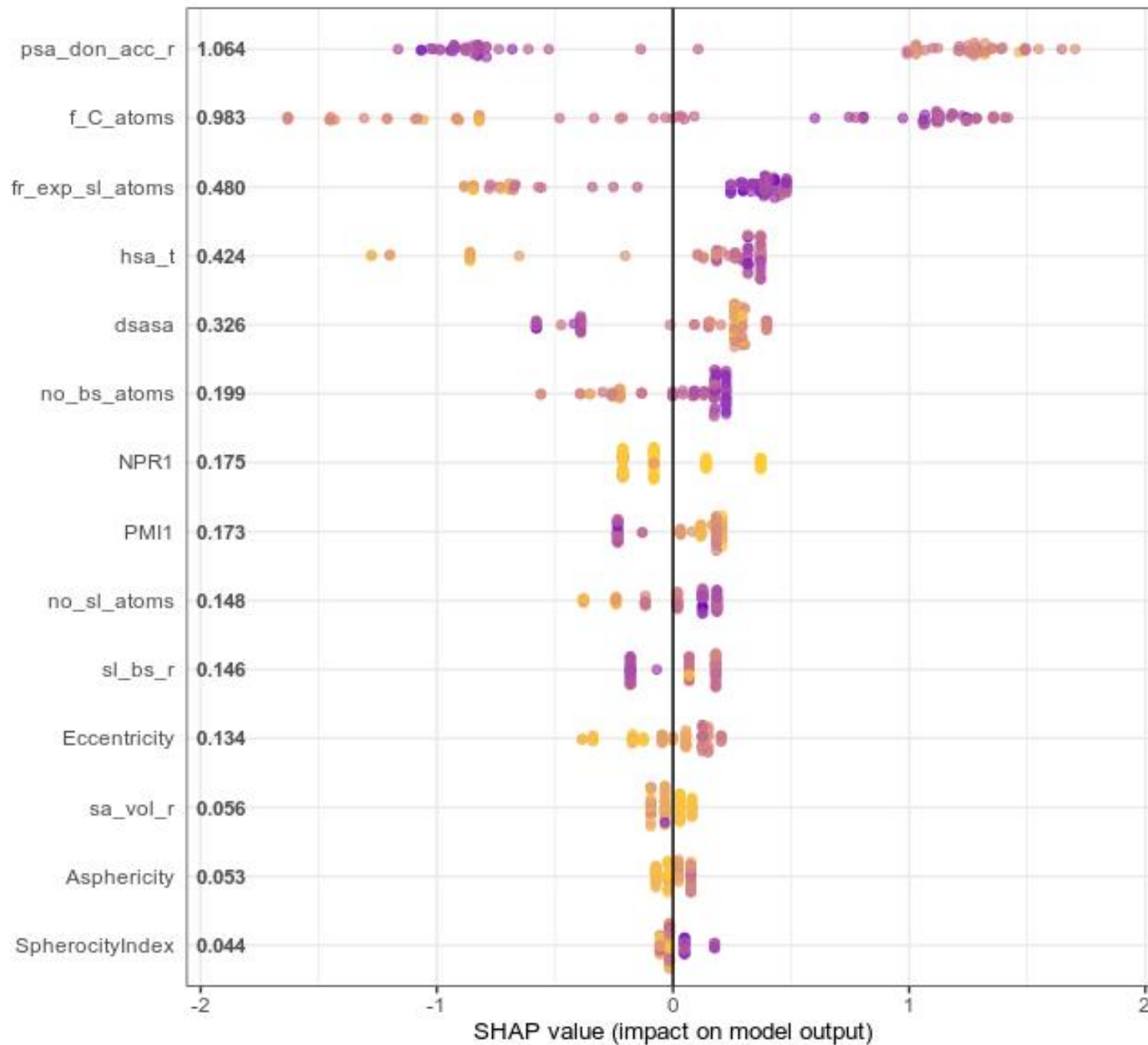
$$D2_{xyz} = us_{xyz} + N_{xyz}$$

- Recalculate SASA

Hydrophobic surface area

- $\Sigma(\Delta(\text{Unblocked-Blocked}))$

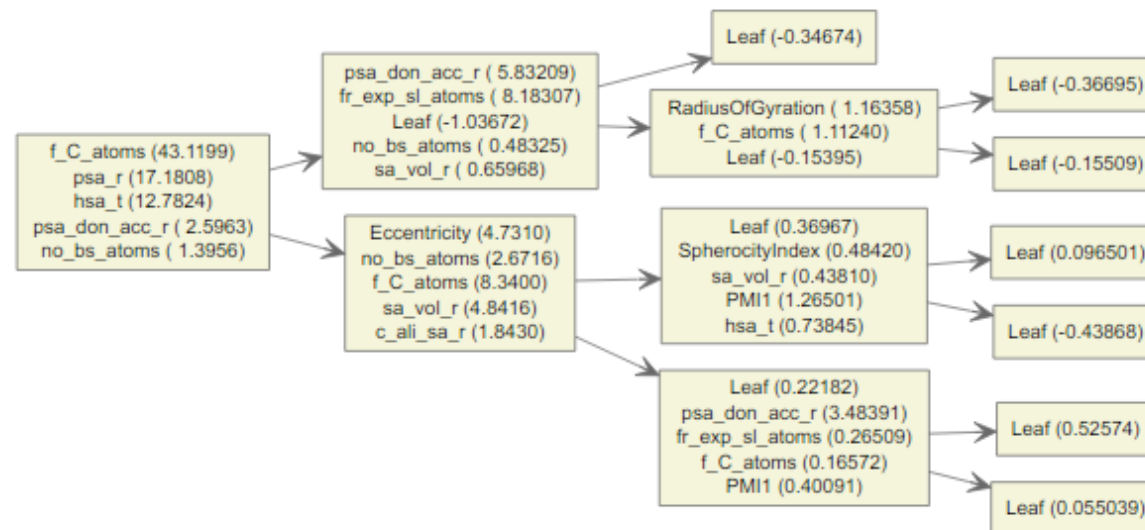


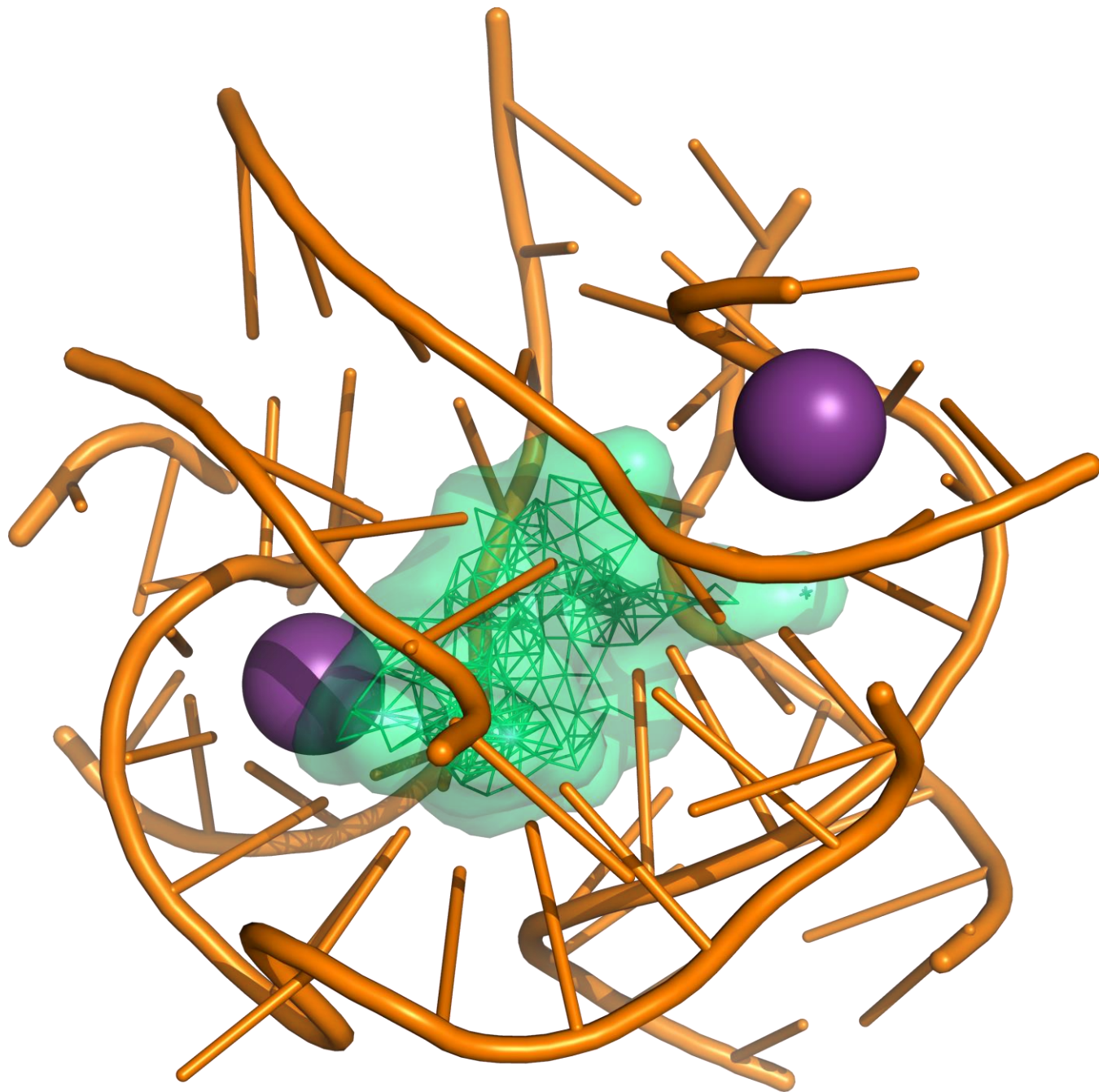


High impact on model
Low impact on model

XGBoost

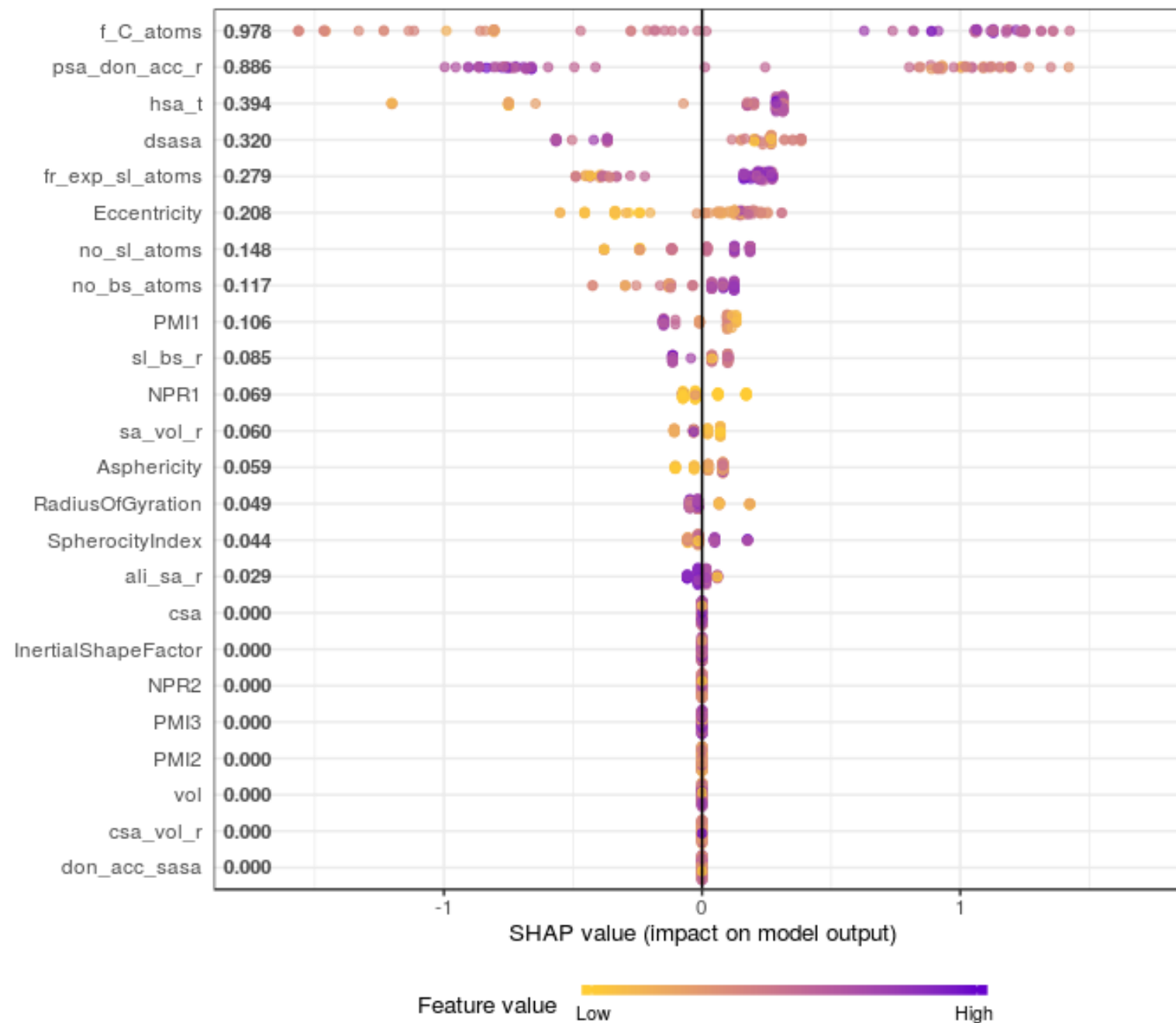
```
[1] train-error:0.133333
[2] train-error:0.120000
[3] train-error:0.053333
[4] train-error:0.040000
[5] train-error:0.040000
[6] train-error:0.013333
[7] train-error:0.026667
[8] train-error:0.013333
[9] train-error:0.013333
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```





Descriptors – Binding site properties

- Shape descriptors:
Contact surface area
Volume (Å³)
Principal Moment of Ratio
et cetera...
- Apolar descriptors:
Hydrophobic surface area
Fraction of carbon atoms
- Polar descriptors:
Total surface area – Apolar surface area
-



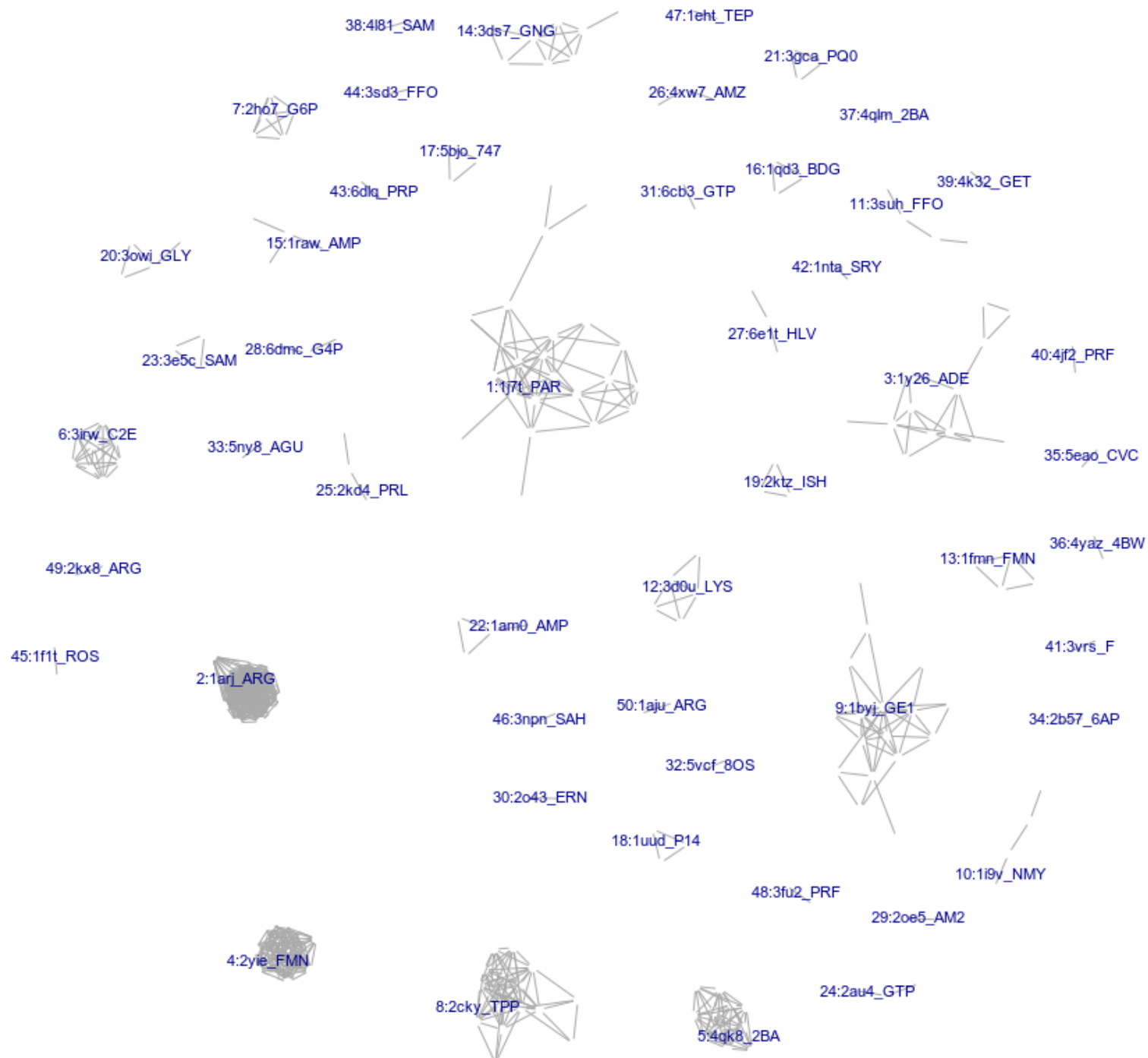
How do you make a prediction software?

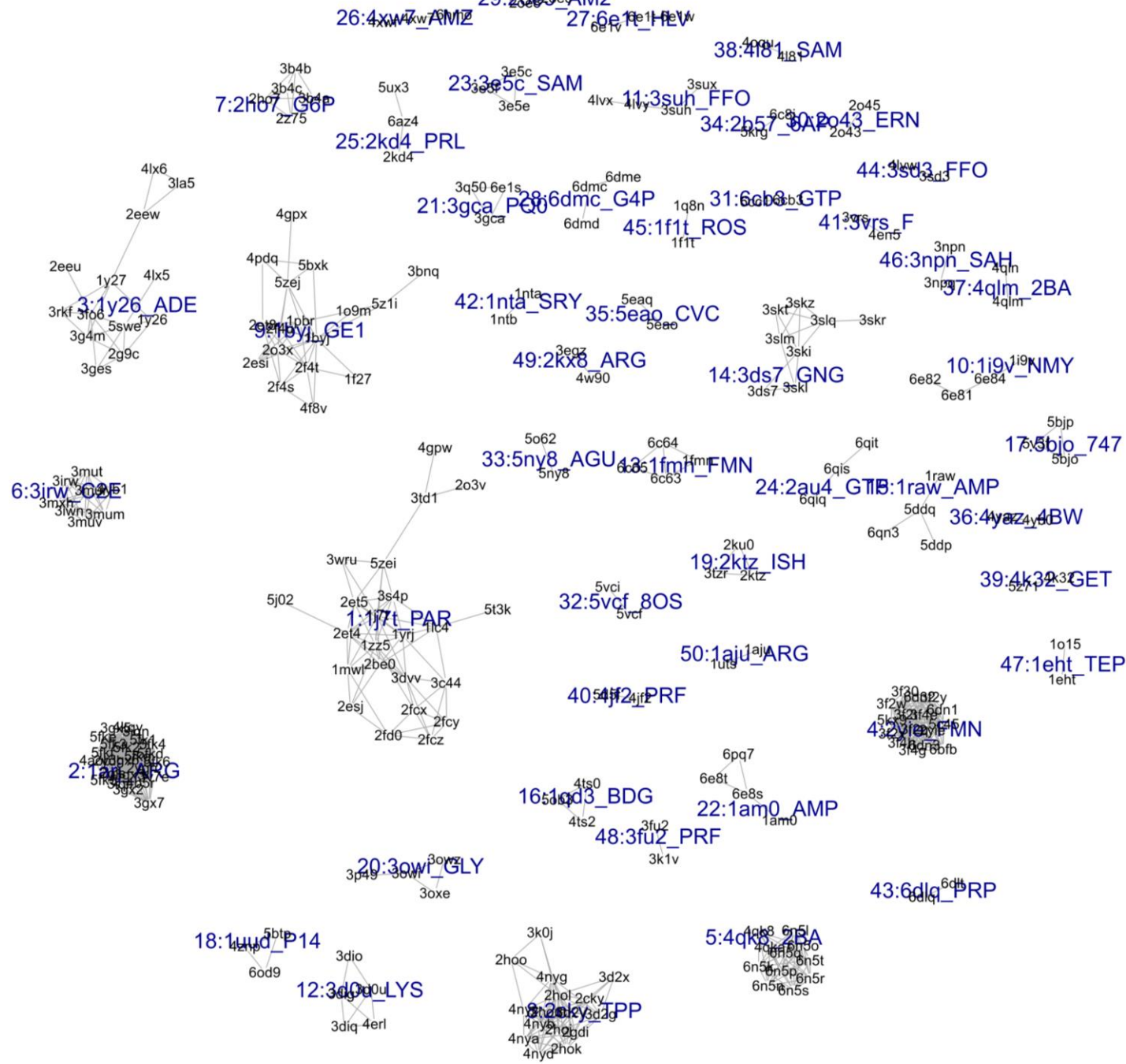
- Training sets!



- Aim: Find out if someone is a Bergenser without asking if they are a Bergenser
- Ask a lot of Bergen and non-Bergen people the same questions
- Two different sets with unique trending answers
- Build a model
- Ask any random Norwegian the same questions and you will discover his/her true Bergenser-status using the model







	Training [druggable / less druggable]		Testing [druggable / less druggable]	
	Drug pred _RN A	Drug Pred 2.0	Drug pred _RN A	Drug Pred 2.0
Accuracy	0.99	0.91	0.89	0.94
Precision	1.00 / 0.97	0.92 / 0.89	0.90 / 0.85	0.95 / 0.93
Recall	0.98 / 1.00	0.94 / 0.86	0.90 / 0.85	0.95 / 0.93

Group no	Head	Comment	Members	Consensus [%]	Outcome
1	1j7t_PAR	Ribosome (aminoglycoside)	28	85.7 / 82.6	ld
2	1arj_ARG	SAM rs (incorrect label)	64	40.6 / 36.0	ld
3	1y26_ADE	Guanine/Purine/Adenine/... rs	13	100 / 100	ld
4	2yie_FMN	FMN rs	36	83.3 / 87.5	d
5	4qk8_2BA	c-di-AMP	13	53.8 / 45.5	ld
6	3irw_C2E	c-di-GMP	14	57.1 / 75.0	d
7	2ho7_G6P	glmS rz	13	7.69 / 42.86	d
8	2cky_TPP	TPP rs	35	14.3 / 62.5	d
9	1byj_GE1	Ribosome (Kanamycin / Gentamycin / etc)	26	92.3 / 92	ld
10	1i9v_NMY	Neomycin + Corn aptamer	4	100 / 100	ld
11	3suh_FFO	THF rs	4	100 / 100	ld
12	3d0u	Lysine rs	25	100 / 100	ld
20	3owi_GLY	Glycine rs	4	100 / 100	ld
21	3gca_PQ0	PreQ1 rs	3	100 / 100	d
28	6dmc_G4P	ppGpp rs	10	60 / 100	d
49	2kx8_ARG	Arginine rs	11	27 / 20	ld

